Large Language Models for Biological Sequences

Do the models learn, or do they just memorize?

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sinc(*i*)

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03/11/26

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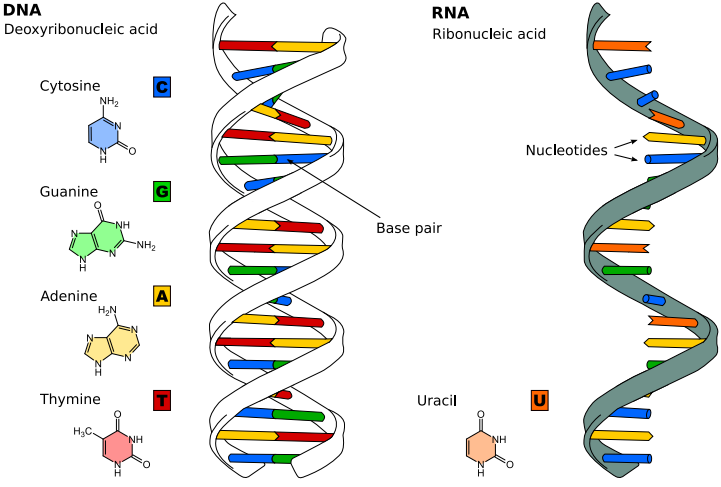
From DNA to RNA



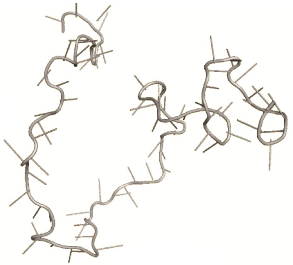
*Adapted from Wikipedia

Structure of RNA: folding

GCGGAUUUAG CUCAGUUGGG AGAGCGCCAG ACUGAAGAUC UGGAGGUCCU GUGUUCGAUC CACAGAAUUC GCACCA

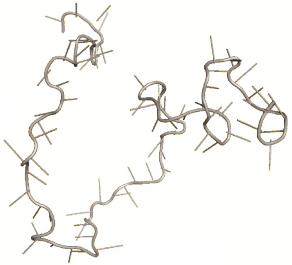
Structure of RNA: folding

GCGGAUUUAG CUCAGUUGGG AGAGCGCCAG ACUGAAGAUC UGGAGGUCCU GUGUUCGAUC CACAGAAUUC GCACCA



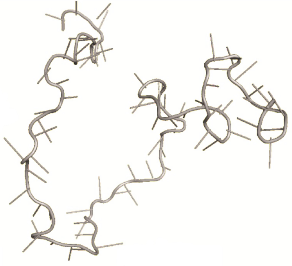
Structure of RNA: folding

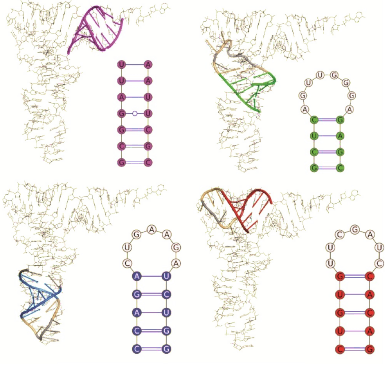
GCGGAUUUAG CUCAGUUGGG AGAGCGCCAG ACUGAAGAUC UGGAGGUCCU GUGUUCGAUC CACAGAAUUC GCACCA



Structure of RNA: folding

GCGGAUUUAG CUCAGUUGGG AGAGCGCCAG ACUGAAGAUC UGGAGGUCCU GUGUUCGAUC CACAGAAUUC GCACCA





Structure of RNA: folding

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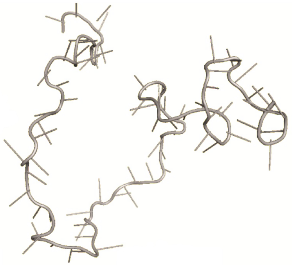
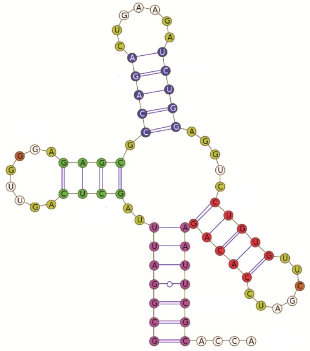
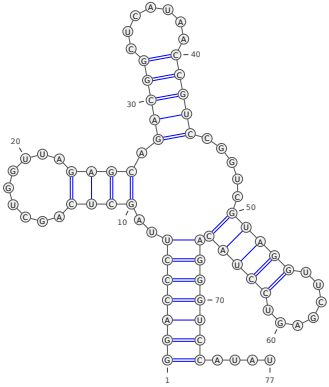
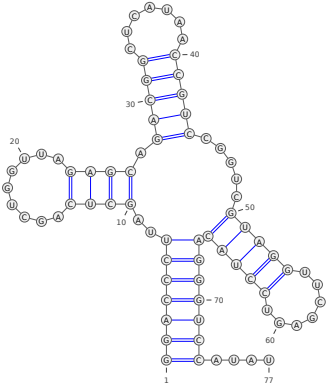


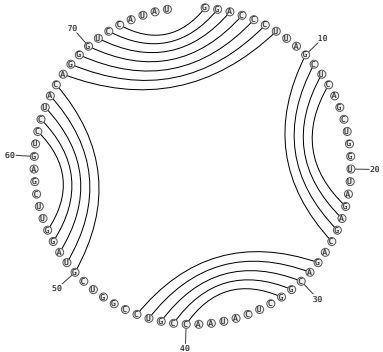


Structure of RNA: alternative representations

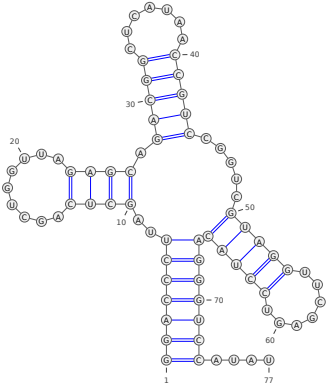


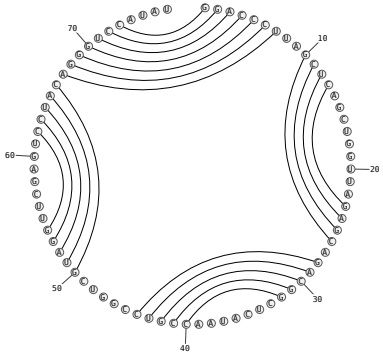
Structure of RNA: alternative representations

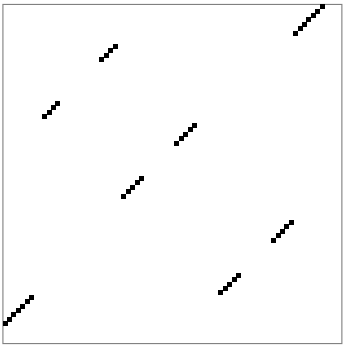




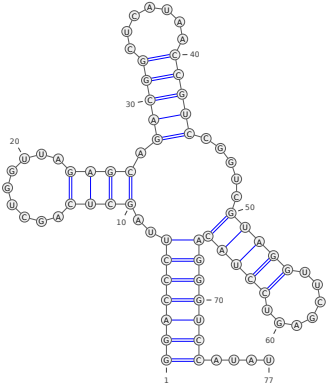
Structure of RNA: alternative representations

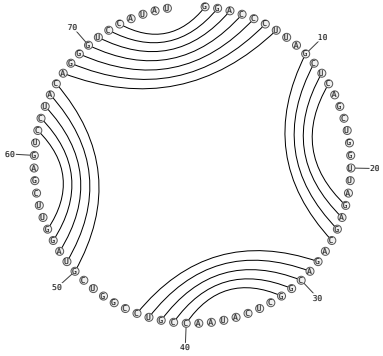


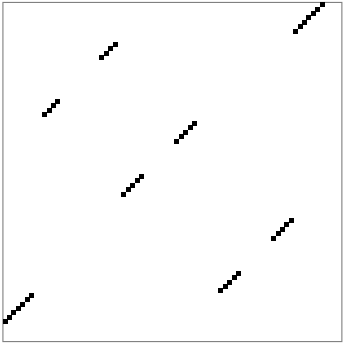




Structure of RNA: alternative representations







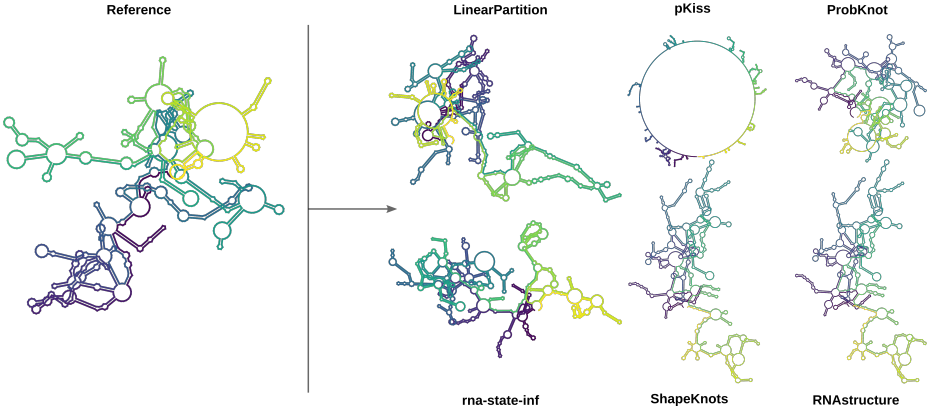
```
GGACCCUUAGCUCAGCUGGUUAGAGCAGACGGCUCAUAACCGUCCGGUCGUAGGUUCGAGUCCUACAGGGUCCAUAU
(((((((..((((.........)))).(((((.......)))))).....(((((.......)))))))))))....
```

Datasets (ArchiveII)

id	sequence	structure	base_pairs
srp_Shig.flex._CP000266	CCGUCAGGUCCGGAAGGAAGCAGCGGUA	((((....(((....)))....))))..	[[1, 26], [2, 25], [3, 24], [4, 23], [9, 18], [10, 17], [11, 16]]
srp_Sino.medi._CP000738	GCCAACCCGGUCAGGUCCGGAAGGAAGCAGCCG	.......(((....(((....)))....)))	[[8, 33], [9, 32], [10, 31], [11, 30], [16, 25], [17, 24], [18, 23]]
srp_Para.deni._CP000489	AACCGGGUCAGGUCCGGAAGGAAGCAGCCCUAA	....(((((....(((....)))....))))...	[[5, 30], [6, 29], [7, 28], [8, 27], [13, 22], [14, 21], [15, 20]]
srp_Lact.lact._AM406671	CGAAGCGUGUCAGGACCUGACGGUAGCAGCACUAAG	(.....(((((....(((....)))....)))))...)	[[1, 36], [7, 32], [8, 31], [9, 30], [10, 29], [15, 24], [16, 23], [17, 22]]
srp_Meth.vann._CP000742	CGAACCCCGGUCAGGCCCGGAAGGGAGCAGCGGUAGG	(.....(((((....(((....)))....)))))..).	[[1, 35], [7, 32], [8, 31], [9, 30], [10, 29], [15, 24], [16, 23], [17, 22]]
srp_Ther.mela._CP000716	GUGAACCGGGUCAGGUCCGGAAGGAGCAGCCCUAAGC	((.....(((((....(((....)))....)))))...))	[[1, 38], [2, 37], [8, 33], [9, 32], [10, 31], [11, 30], [16, 25], [17, 24], [1
srp_Rhod.spha._CP000143	UCGCCAACCCGGUCAGGUCCGGAAGGAAGCAGCCGUAA	((((.....(((((....(((....)))....)))))...))))	[[1, 41], [2, 40], [3, 39], [10, 35], [11, 34], [12, 33], [13, 32], [18, 27], [
srp_Acin.spec._CR543861	GCCGCGAACCCCGCCAGGCCCGGAAGGGAGCAACGGUA	((((.....(((.....(((....)))....)))..))))	[[1, 43], [2, 42], [3, 41], [4, 40], [5, 39], [11, 36], [12, 35], [13, 34], [19
srp_Sacc.eryt._AM420293	CUGAACCCCCGCAGGGCCGGAAGGCAGCAAGGGUAGGU	((.....((((.....(((....)))....)))).)).	[[1, 37], [2, 36], [8, 33], [9, 32], [10, 31], [16, 25], [17, 24], [18, 23]]
srp_Baci.subt._X06803	GAAUUCAUGAACCAUGUCAGGUCCGGAAGGAAGCAGCA	(.((((((.....(((((....)))))...)))))).)	[[1, 52], [2, 51], [4, 47], [5, 46], [6, 45], [7, 44], [8, 43], [14, 39], [15,
srp_Myco.spec._CP000518	UGAACUCCCCCAGGGCCGGAAGGCAGCAAGGGUAAGCG	(.....(((.....(((....)))....)))..)..).	[[1, 35], [7, 32], [8, 31], [9, 30], [15, 24], [16, 23], [17, 22]]
tRNA_tdbR00000398-Ascaris	GACAAAUGUUUUCAGGUCUUCUAAAUCUGUUUUGGAGA	((((((.....(((((.....)))))....((((.....	[[1, 50], [2, 49], [3, 48], [4, 47], [5, 46], [6, 45], [7, 44], [13, 29], [14,
srp_Shig.flex._AE016979	GUGAACCCGGUCAGGCCCGGAAGGGAGCAGCCGAAGCG	((.....(((((....(((....)))....)))))...)).	[[1, 37], [2, 36], [8, 33], [9, 32], [10, 31], [11, 30], [16, 25], [17, 24], [1
tRNA_tdbR00000363-Ascaris	GACAAAUGUUUUCAGGUCUUCUAAAUCUGUUUUGGAGA	((((((.....(((((.....)))))....)))))..).	[[1, 50], [2, 49], [3, 48], [4, 47], [5, 46], [6, 45], [7, 44], [20, 36], [21,
srp_Lact.brev._CP000416	GCAAGGUGGGCUUACGAAUCGUGUCAGGUCCGGAAGG	...........(..(((.....(((....)))....)))	[[12, 55], [15, 52], [16, 51], [22, 47], [23, 46], [24, 45], [25, 44], [30, 39]
srp_Bdel.bact._BX842656	UGGGCCCCACGCAACGGAAGGUUACGAACUCCGUCAGG	..........(((((.....(((....)))....)))))	[[31, 56], [32, 55], [33, 54], [34, 53], [39, 48], [40, 47], [41, 46]]
srp_Stre.pyog._AM295007	GGAGCAACACUUGUGCGUGAAGUGGGUCAGGGGAGGAA	..............(((((.....(((....)))....)))))	[[16, 55], [17, 54], [18, 53], [24, 49], [25, 48], [26, 47], [27, 46], [32, 41]
tRNA_tdbR00000364-Ascaris	GGACGUUAAAUAGAUAAGCUAUGCCUAGUUACGGGCUG	(((((.(.(((((.....)))))).(((((.....)))))	[[1, 53], [2, 52], [3, 51], [4, 50], [5, 49], [7, 47], [10, 22], [11, 21], [12,
srp_Magn.magn._AP007255	UCGCCAACCCGGUCAGGUCCGGAAGGAAGCAGCCGUAA	((((.....(((((....(((....)))....)))))...))))	[[1, 41], [2, 40], [3, 39], [10, 35], [11, 34], [12, 33], [13, 32], [18, 27], [
tRNA_tdbR00000592-Ascaris	UAUACUUUGAGUUUAGGUAAGAAUAUUUAUUUUUGGUGUAA	(((((.....(((((.....)))))....(((((.....	[[1, 54], [2, 53], [3, 52], [4, 51], [5, 50], [6, 49], [10, 22], [11, 21], [12,
srp_Pedi.pent._CP000422	AGGCAAGGUUGGACUUAGCAAUUGUGUCAGGUCCGGAA	...........(.(((.....(((....)))....))).)	[[14, 57], [16, 55], [17, 54], [18, 53], [24, 49], [25, 48], [26, 47], [27, 46]
srp_Myco.bovi._BX248347	UGAACUCCCCCAGGGCCGGAAGGGCAGCAAGGGUCAAUG	((.....((((.....(((....)))....)))).)).	[[1, 36], [7, 32], [8, 31], [9, 30], [15, 24], [16, 23], [17, 22]]
tRNA_tdbR00000593-Ascaris	ACAGAUUUAAGUUAAGUUUAAACUCUUGGUUUUCAAAA	(((((.(..(((((.....)))))..(((((.....	[[1, 56], [2, 55], [3, 54], [4, 53], [5, 52], [7, 50], [10, 24], [11, 23], [12,
tRNA_tdbR00000590-Ascaris	GAGAUAUUAGUAUAAAUUUUUUGUAUAUUUCUUUUUCG	(((((.(..(((((.....)))))..(((((.....	[[1, 58], [2, 57], [3, 56], [4, 55], [5, 54], [6, 53], [10, 26], [11, 25], [12,
srp_Sili.spec._CP000377	GUCGGGCGGACGCAGCCUUCGCCAACCCGGUCAGGUCC	((((.....(((((.....)))))....(((((.....	[[18, 60], [19, 59], [20, 58], [21, 57], [28, 53], [29, 52], [30, 51], [31, 50]
tRNA_tdbR00000078-Ascaris	ACUCUGUUAGUUUAUGUUUUAAAAUAUGACUUUGAAGA	((((.....(((((.....)))))....(((((.....	[[1, 58], [2, 57], [3, 56], [4, 55], [5, 54], [10, 25], [11, 24], [12, 23], [13
srp_Coxi.burn._AE016828	GAACCCCGCAGGCCCGGAAGGGAGCAACGGUAGUAGU	.....(((.....(((....)))....))).....(((.	[[6, 31], [7, 30], [8, 29], [14, 23], [15, 22], [16, 21]]
srp_Heli.annu._AJ828095	GGGCAAUGCGUUAGGCUGGUUUCACAGAGCUGCGAAAA	((.....(.(((....(((....)))....))).)..).	[[1, 40], [2, 39], [7, 36], [9, 34], [10, 33], [11, 32], [16, 27], [17, 26], [1
srp_Heli.exil._EE659686	AAUGCGUUAGGCUGGUUUCACAGAGCUGCGAGAACUCC	..(.((((.....(((....)))....))).)).).....	[[3, 32], [5, 30], [6, 29], [7, 28], [12, 23], [13, 22], [14, 21]]
tRNA_tdbR00000282-Ascaris	AAUAAGAUAGGAUAAGUUGAGUCUGUGAGGUUCAUACC	((((((.....(((((.....)))))....(((((.....	[[1, 60], [2, 59], [3, 58], [4, 57], [5, 56], [6, 55], [7, 54], [10, 24], [11,
tRNA_tdbR00000591-Ascaris	GGGGUGUUAACUUAAGUUUAAAGUGUUAGAUUUUUAAU	((((((.(.(((((.....)))))..(((((.....	[[1, 61], [2, 60], [3, 59], [4, 58], [5, 57], [6, 56], [7, 55], [10, 24], [11,
srp_Synt.wolf._CP000448	CCGUGCUAGAUGGGGAGGUAGCGGUGCCGUCGUAACCUG	........(((((.(.(...).)))))....(((((...	[[9, 58], [10, 57], [11, 56], [12, 30], [13, 29], [14, 28], [15, 27], [17, 25],
srp_Gynu.aura._SDB-109565	ACCGGGCUUGGCAACGCUUGCCUAUAACUCAAGUGAGG	(((((((.(.....(.....)).)))))....(((((...	[[1, 40], [2, 39], [3, 38], [4, 23], [5, 22], [6, 21], [7, 20], [10, 17],

Complex structures

Long sequences $\Rightarrow$ complex structures $\Rightarrow$ wrong predictions



RNA folding: MFE and dynamic programming

Free energy parameters for hairpin loop formation

Parameter (number of nt or sequence)	ΔG°_{37}, kcal/mol
$\Delta G^\circ_{37\ \text{initiation}}(3)$	5.4 ± 0.2
$\Delta G^\circ_{37\ \text{initiation}}(4)$	5.6 ± 0.1
$\Delta G^\circ_{37\ \text{initiation}}(5)$	5.7 ± 0.2
$\Delta G^\circ_{37\ \text{initiation}}(6)$	5.4 ± 0.1
$\Delta G^\circ_{37\ \text{initiation}}(7)$	6.0 ± 0.2
$\Delta G^\circ_{37\ \text{initiation}}(8)$	5.5 ± 0.2
$\Delta G^\circ_{37\ \text{initiation}}(9)$	6.4 ± 0.2
$\Delta G^\circ_{37\ \text{bonus}}$(UU or GA first mismatch but not AG)	-0.9 ± 0.1
$\Delta G^\circ_{37\ \text{bonus}}$(GG first mismatch)	-0.8 ± 0.3
$\Delta G^\circ_{37\ \text{bonus}}$(special G-U closure)	-2.2 ± 0.2
$\Delta G^\circ_{37\ \text{penalty}}(C_3$ loop)	1.5 ± 0.5
$\Delta G^\circ_{37\ \text{penalty}}(C_n$ loop), A	0.3 ± 0.1
$\Delta G^\circ_{37\ \text{penalty}}(C_n$ loop), B	1.6 ± 0.9

—○ Mathews DH, Disney MD, Childs-Disney JL, Schroeder SJ, Zuker M, Turner DH. Incorporating chemical modification constraints into a dynamic programming algorithm for prediction of RNA secondary structure. PNAS 2004;101(19):7287-7292.

RNA folding: MFE and dynamic programming

Lookup table for unstable triloops and stable tetraloops and hexaloops

Hairpin	Ref(s).	$\Delta G^\circ_{37\,\mathrm{loop}}$, kcal/mol
C**AAC**G	87	6.8
G**UUA**C	87	6.9
C**AACG**G	48	5.5
C**CAAG**G	48	3.3
C**CACG**G	48	3.7
C**CCAG**G	48	3.4
C**CGAG**G	48	3.5
C**CGCG**G	48	3.6
C**CUAG**G	48	3.7
C**CUCG**G	48	2.5
C**UAAG**G	48	3.6
C**UACG**G	47, 50	2.8
C**UCAG**G	48	3.7
C**UCCG**G	47	2.7
C**UGCG**G	47	2.8
C**UUAG**G	48	3.5
C**UUCG**G	47, 50	3.7
A**CAGUAC**U	86	2.8
A**CAGUGC**U	86	2.9
A**CAGUGA**U	86	3.6
A**CAGUUC**U	86	1.8

RNA folding: MFE and dynamic programming

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C**CACG**G	48	3.7
C**CCAG**G	48	3.4
C**CGAG**G	48	3.5
C**CGCG**G	48	3.6
C**CUAG**G	48	3.7
C**CUCG**G	48	2.5
C**UAAG**G	48	3.6
C**UACG**G	47, 50	2.8
C**UCAG**G	48	3.7
C**UCCG**G	47	2.7
C**UGCG**G	47	2.8
C**UUAG**G	48	3.5
C**UUCG**G	47, 50	3.7
A**CAGUAC**U	86	2.8
A**CAGUGC**U	86	2.9
A**CAGUGA**U	86	3.6
A**CAGUUC**U	86	1.8

$$V(i,j) = V_{stack}(i,j) + V_{hairpin}(i,j) + V_{internal}(i,j) + V_{multibranch}(i,j)$$

$$V_{hairpin}(i,j) = e^{-\Delta G(hairpin)/RT}$$

RNA folding: MFE and dynamic programming

Lookup table for unstable triloops and stable tetraloops and hexaloops

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C**CAAG**G	48	3.3
C**CACG**G	48	3.7
C**CCAG**G	48	3.4
C**CGAG**G	48	3.5
C**CGCG**G	48	3.6
C**CUAG**G	48	3.7
C**CUCG**G	48	2.5
C**UAAG**G	48	3.6
C**UACG**G	47, 50	2.8
C**UCAG**G	48	3.7
C**UCCG**G	47	2.7
C**UGCG**G	47	2.8
C**UUAG**G	48	3.5
C**UUCG**G	47, 50	3.7
A**CAGUAC**U	86	2.8
A**CAGUGC**U	86	2.9
A**CAGUGA**U	86	3.6
A**CAGUUC**U	86	1.8

$$V(i,j) = V_{stack}(i,j) + V_{hairpin}(i,j) + V_{internal}(i,j) + V_{multibranch}(i,j)$$

$$V_{hairpin}(i,j) = e^{-\Delta G(hairpin)/RT}$$

$$V_{stack}(i,j) = e^{-\Delta G(stack)/RT} \times V(i+1, j-1)$$

RNA folding: MFE and dynamic programming

Lookup table for unstable triloops and stable tetraloops and hexaloops

Hairpin	Ref(s).	$\Delta G°_{37\ loop}$, kcal/mol
C**AAC**G	87	6.8
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C**AACG**G	48	5.5
C**CAAG**G	48	3.3
C**CACG**G	48	3.7
C**CCAG**G	48	3.4
C**CGAG**G	48	3.5
C**CGCG**G	48	3.6
C**CUAG**G	48	3.7
C**CUCG**G	48	2.5
C**UAAG**G	48	3.6
C**UACG**G	47, 50	2.8
C**UCAG**G	48	3.7
C**UCCG**G	47	2.7
C**UGCG**G	47	2.8
C**UUAG**G	48	3.5
C**UUCG**G	47, 50	3.7
A**CAGUAC**U	86	2.8
A**CAGUGC**U	86	2.9
A**CAGUGA**U	86	3.6
A**CAGUUC**U	86	1.8

$$V(i,j) = V_{stack}(i,j) + V_{hairpin}(i,j) + V_{internal}(i,j) + V_{multibranch}(i,j)$$

$$V_{hairpin}(i,j) = e^{-\Delta G(hairpin)/RT}$$

$$V_{stack}(i,j) = e^{-\Delta G(stack)/RT} \times V(i+1, j-1)$$

$$V_{internal}(i,j) = \Sigma\Sigma\, V(i', j') \times e^{-\Delta G(internal)/RT}$$

RNA folding: MFE and dynamic programming

Lookup table for unstable triloops and stable tetraloops and hexaloops

Hairpin	Ref(s).	$\Delta G°_{37\ loop}$, kcal/mol
CAACG	87	6.8
GUUAC	87	6.9
CAACGG	48	5.5
CCAAGG	48	3.3
CCACGG	48	3.7
CCCAGG	48	3.4
CCGAGG	48	3.5
CCGCGG	48	3.6
CCUAGG	48	3.7
CCUCGG	48	2.5
CUAAGG	48	3.6
CUACGG	47, 50	2.8
CUCAGG	48	3.7
CUCCGG	47	2.7
CUGCGG	47	2.8
CUUAGG	48	3.5
CUUCGG	47, 50	3.7
ACAGUACU	86	2.8
ACAGUGCU	86	2.9
ACAGUGAU	86	3.6
ACAGUUCU	86	1.8

$$V(i,j) = V_{stack}(i,j) + V_{hairpin}(i,j) + V_{internal}(i,j) + V_{multibranch}(i,j)$$

$$V_{hairpin}(i,j) = e^{-\Delta G(hairpin)/RT}$$

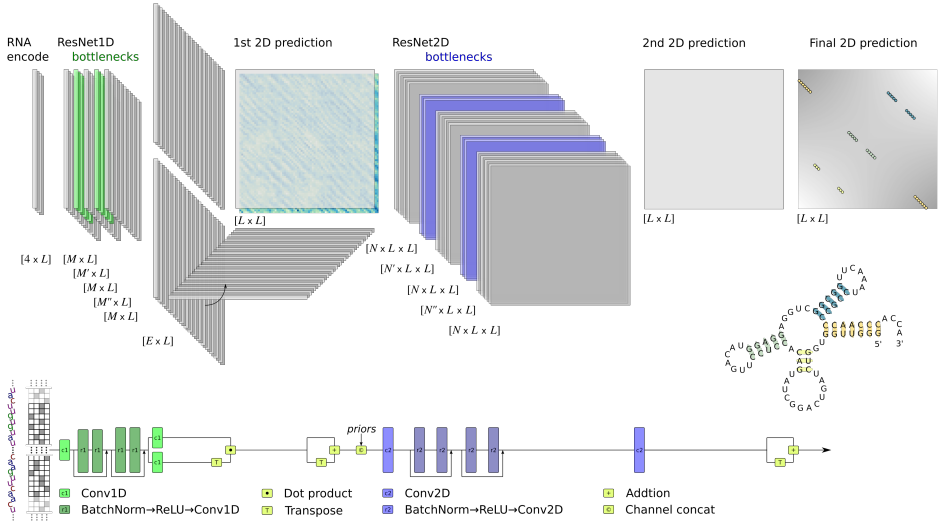
$$V_{stack}(i,j) = e^{-\Delta G(stack)/RT} \times V(i+1, j-1)$$

$$V_{internal}(i,j) = \Sigma\Sigma\, V(i', j') \times e^{-\Delta G(internal)/RT}$$

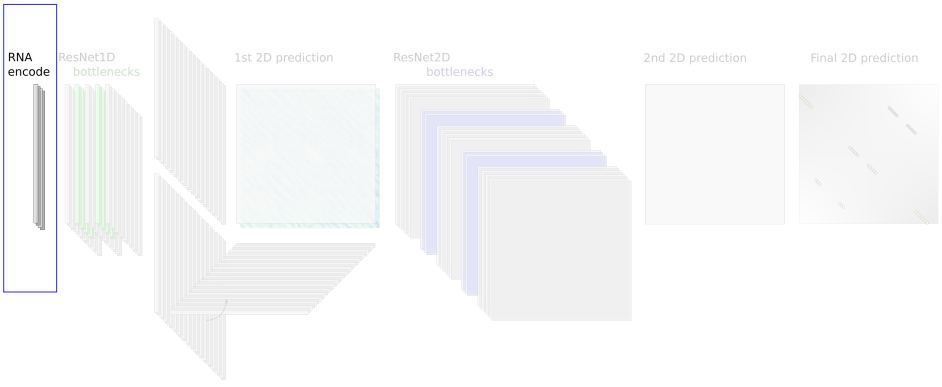
$$\begin{aligned} V_{multibranch}(i,j) = {} & WMB(i+1, j-1) \times (a') \times (c') \\ & + e^{-\Delta G(3' \text{ dangle})/RT}\, xWMB(i+2, j-1) \times (a') \times (b') \times (c') \\ & + e^{-\Delta G(5' \text{ dangle})/RT}\, xWMB(i+1, j-2) \times (a') \times (b') \times (c') \\ & + e^{-\Delta G(\text{terminal mismatch})/RT}\, xWMB(i+2, j-2) \times (a') \times (b')^2 \\ & + \Sigma\, e^{-\Delta G(\text{coaxial stacking})/RT} \times V(i+1,k) \times (W(k+1, j-1) \\ & \quad + WMB(k+1, j-1)) \times (a') \times (c')^2 \\ & + \Sigma\, e^{-\Delta G(\text{coaxial stacking})/RT} \times V(k, j-1) \times (W(i+1, k-1) \\ & \quad + WMB(i+1, k-1)) \times (a') \times (c')^2 \\ & + \Sigma\, e^{-\Delta G(\text{coaxial stacking})/RT} \times V(i+2,k) \times (W(k+2, j-1) \\ & \quad + WMB(k+2, j-1)) \times (a') \times (b')^2 \times (c')^2 \\ & + \Sigma\, e^{-\Delta G(\text{coaxial stacking})/RT} \times V(i+2,k) \times (W(k+1, j-2) \\ & \quad + WMB(k+1, j-2)) \times (a') \times (b')^2 \times (c')^2 \\ & + \Sigma\, e^{-\Delta G(\text{coaxial stacking})/RT} \times V(k, j-2) \times (W(i+1, k-2) \\ & \quad + WMB(i+1, k-2)) \times (a') \times (b')^2 \times (c')^2 \\ & + \Sigma\, e^{-\Delta G(\text{coaxial stacking})/RT} \times V(k, j-2) \times (W(i+2, k-1) \\ & + WMB(i+2, k-1)) \times (a') \times (b')^2 \times (c')^2 \end{aligned}$$

sincFold

—o Bugnon LA, Di Persia LE, Gerard MF, Raad J, Prochetto S, Fenoy E, Chorostecki U, Ariel F, Stegmayer G, Milone DH. sincFold: end-to-end learning of short- and long-range interactions in RNA secondary structure. Briefings in Bioinformatics 2024;25(4). doi:10.1093/bib/bbae271

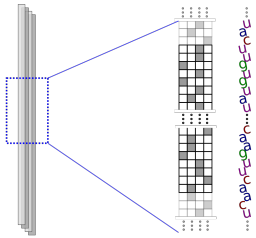


sincFold: input encoding



sincFold: input encoding

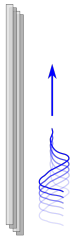
RNA
encode



$[4 \times L]$

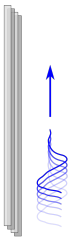
sincFold: 1D convolutions

RNA
encode



sincFold: 1D convolutions

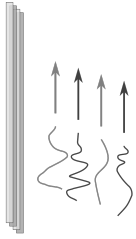
RNA
encode



$$y[n] = \sum_k w[n-k]x[k]$$

sincFold: 1D convolutions

RNA
encode

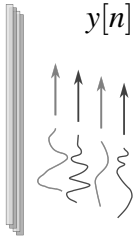


$[4 \times L]$

sincFold: 1D convolutions

RNA
encode

$$y[n] = \sum_i \sum_k w_i[n-k] x_i[k]$$

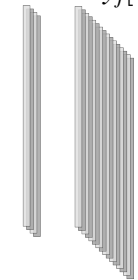


$[4 \times L]$

sincFold: 1D convolutions

RNA encode

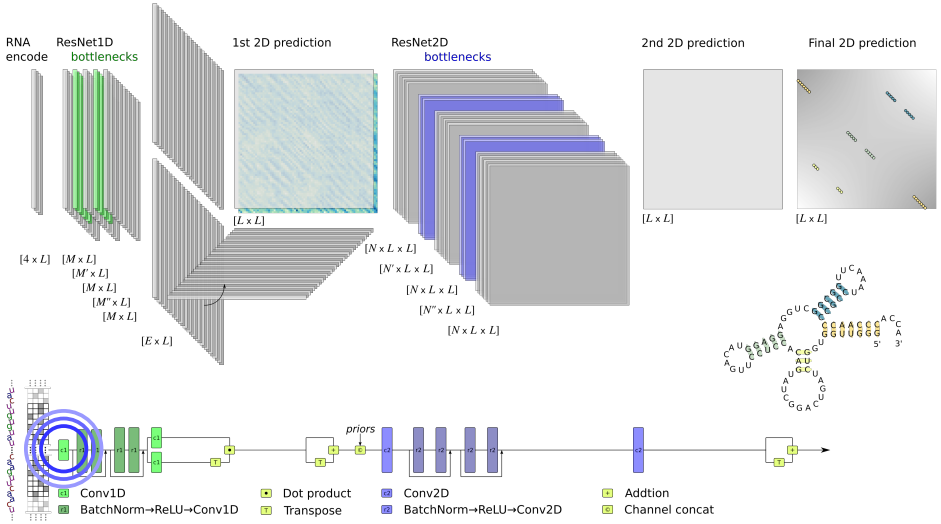
$$y_j[n] = \sum_i \sum_k w_{ij}[n-k]x_i[k]$$



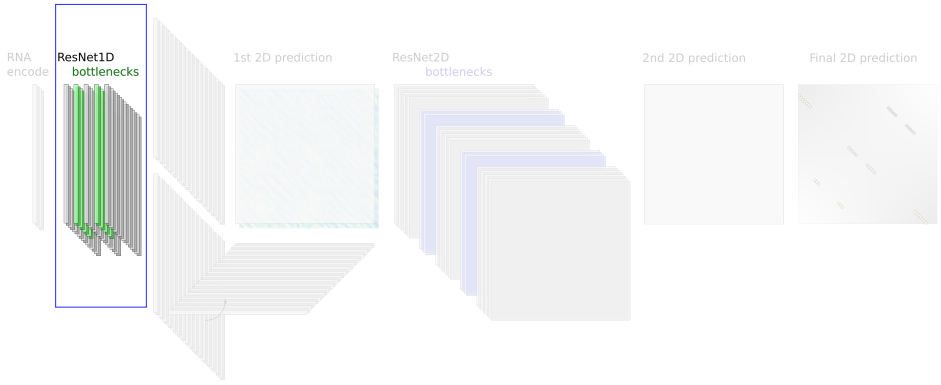
$[4 \times L]$ $[M \times L]$

Conv1D

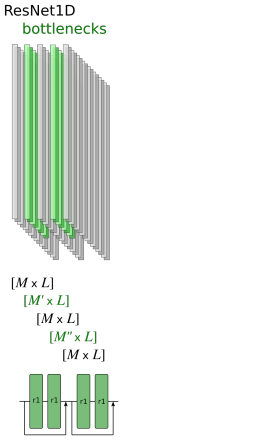
sincFold: first convolutional block



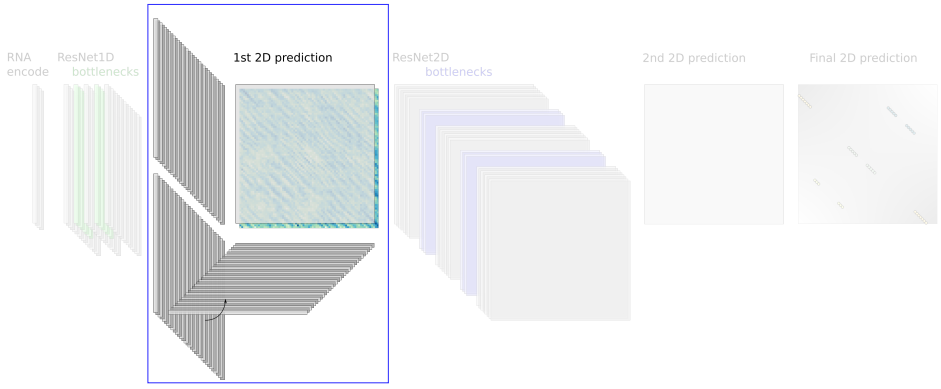
sincFold: 1D convolutional bottlenecks



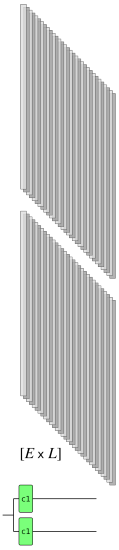
sincFold: 1D convolutional bottlenecks



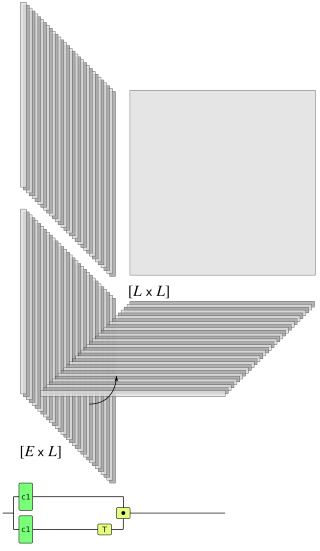
sincFold: 1D embeddings → 2D representation



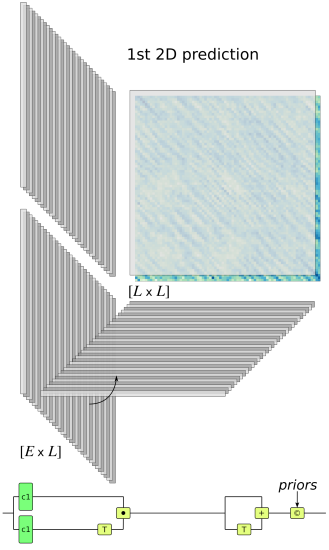
sincFold: 1D → 2D



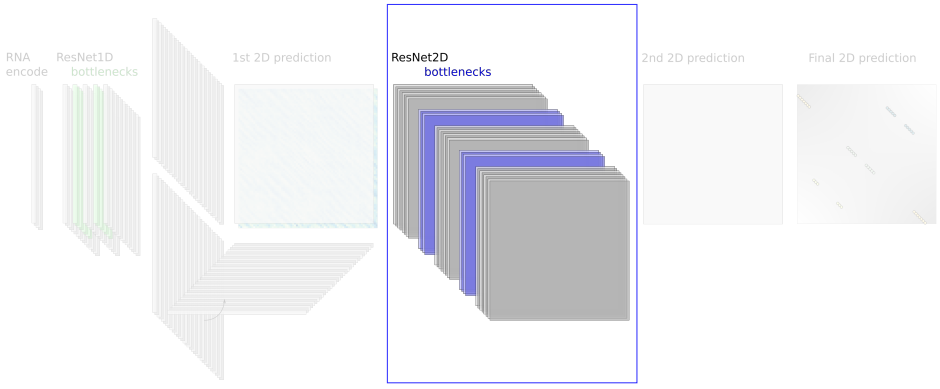
sincFold: 1D $\rightarrow$ 2D



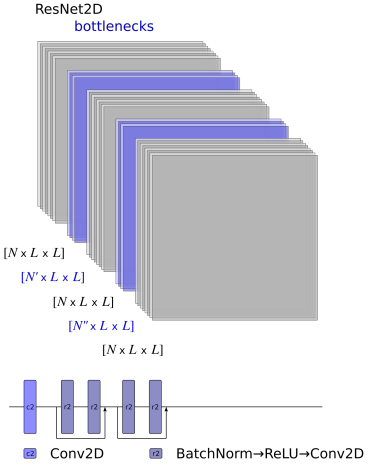
sincFold: 2D symmetrization and priors



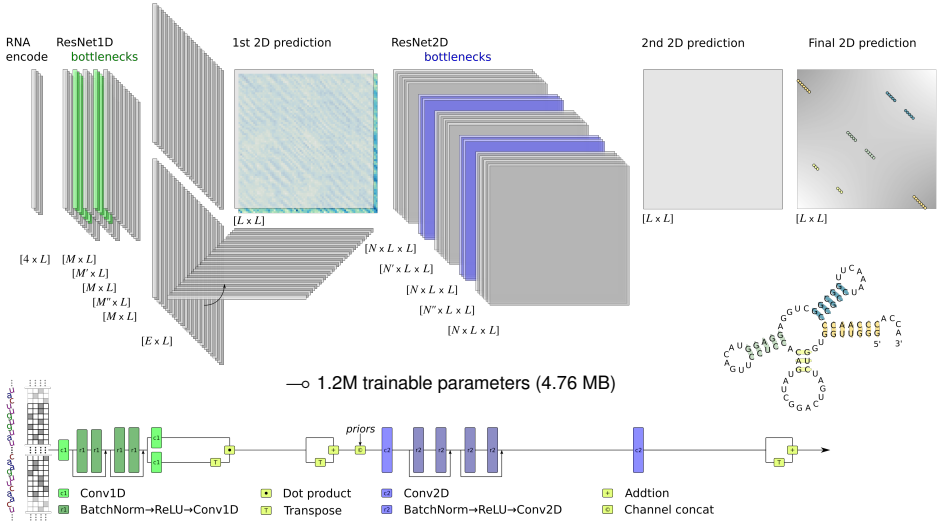
sincFold: 2D convolutional bottlenecks



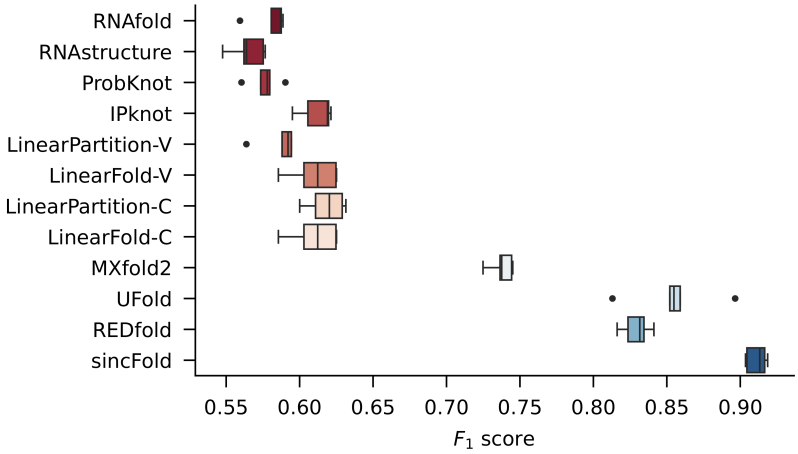
sincFold: 2D convolutional bottlenecks



sincFold: output connection matrix



sincFold: k-fold on ArchiveII



Same problem... now with LLMs

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  - —∘ 120-dim, 76.2k seqs, RNAcentral
  - ↬ transformer (6), ∼509k params
  - ↬ MLM 15%, plus Rfam seed alignments

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  - —∘ 640-dim, 23.7M seqs, RNAcentral
  - ↬ transformer (12), ∼100M params
  - ↬ MLM 15%, original BERT

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- RNA-MSM (2024)
  - —∘ 768-dim, 3.1M seqs, Rfam
  - ↬ transformer (12), ∼96M params
  - ↬ MLM 20% over MSAs, AlphaFold2-inspired

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- ERNIE-RNA (2024)
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- RiNALMo (2024)
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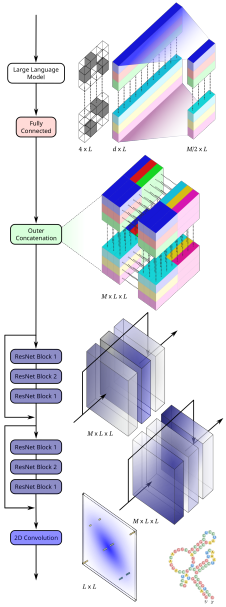
All pre-trained with masked tokens

—∘ all frozen: feature extractors only

The prediction model



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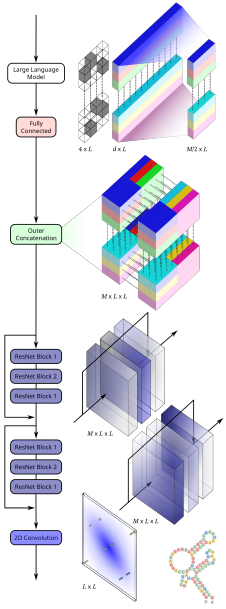

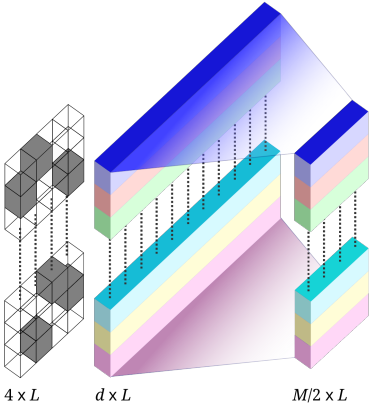

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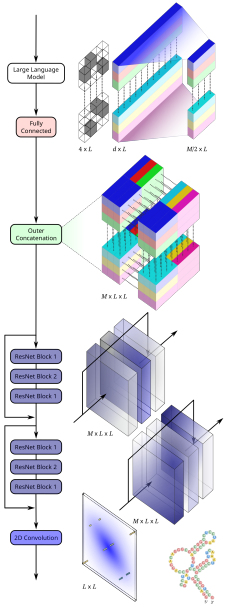

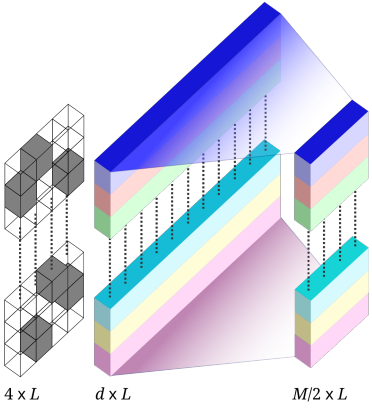

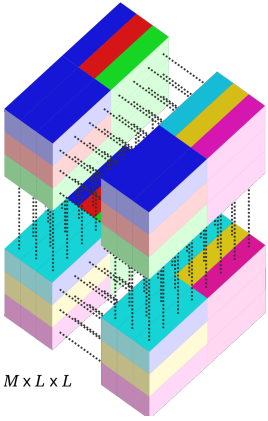

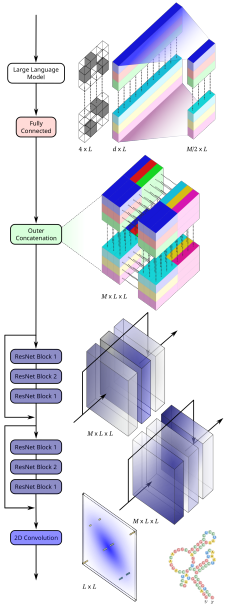

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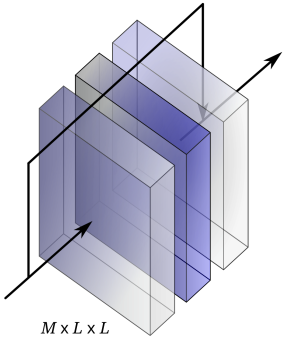
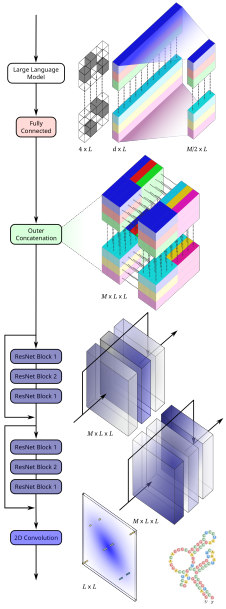
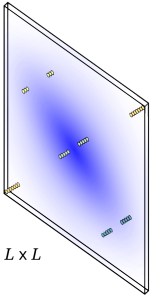


The prediction model





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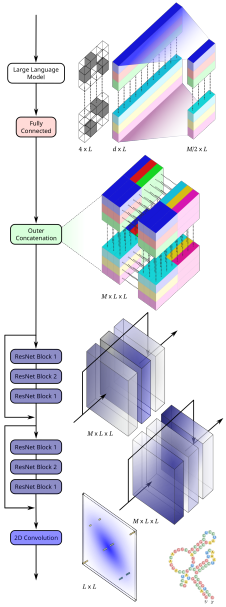
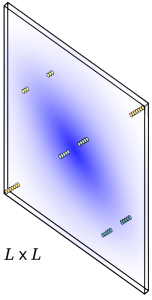
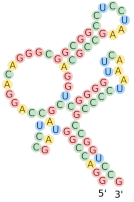
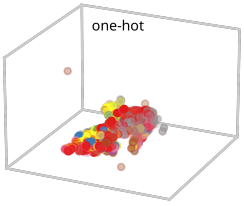
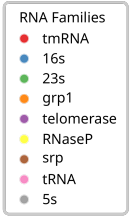






What the embeddings look like (UMAP)





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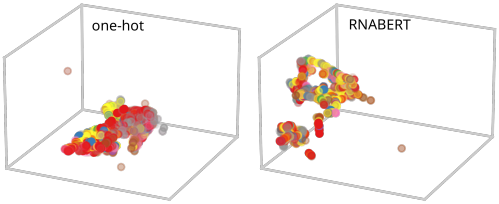

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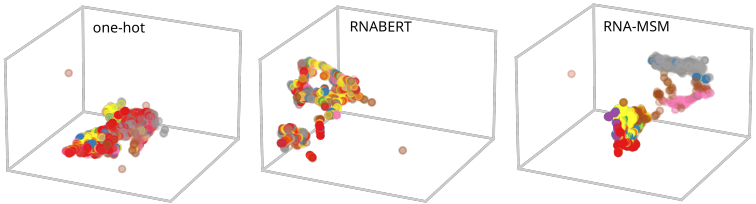

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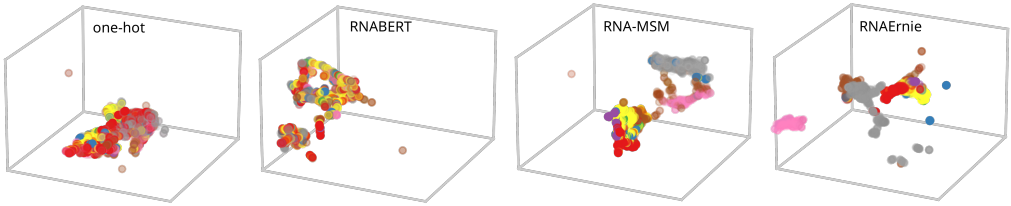

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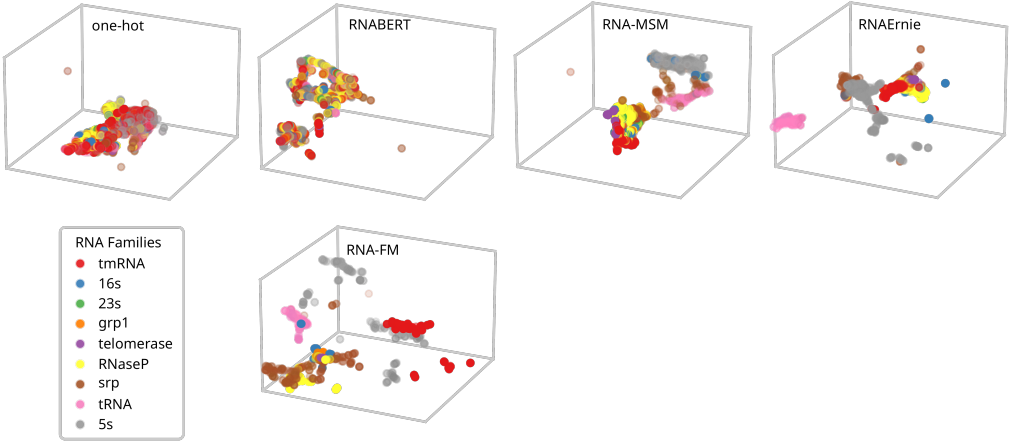

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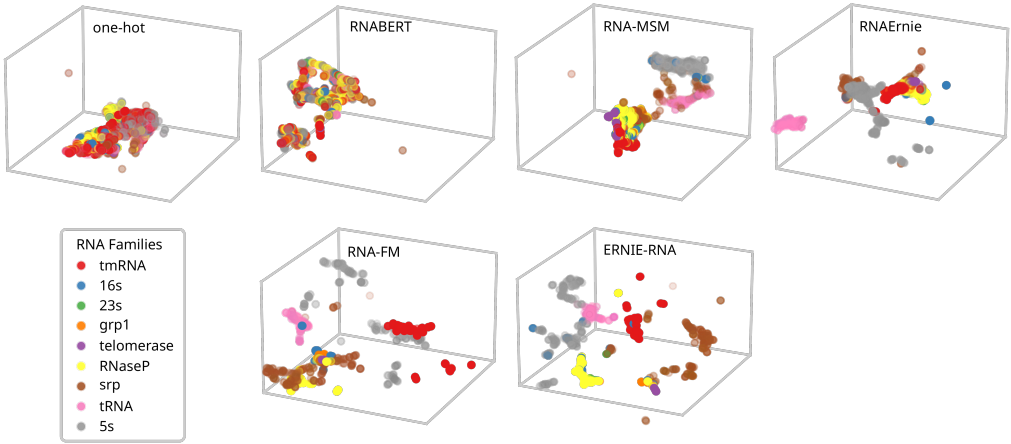

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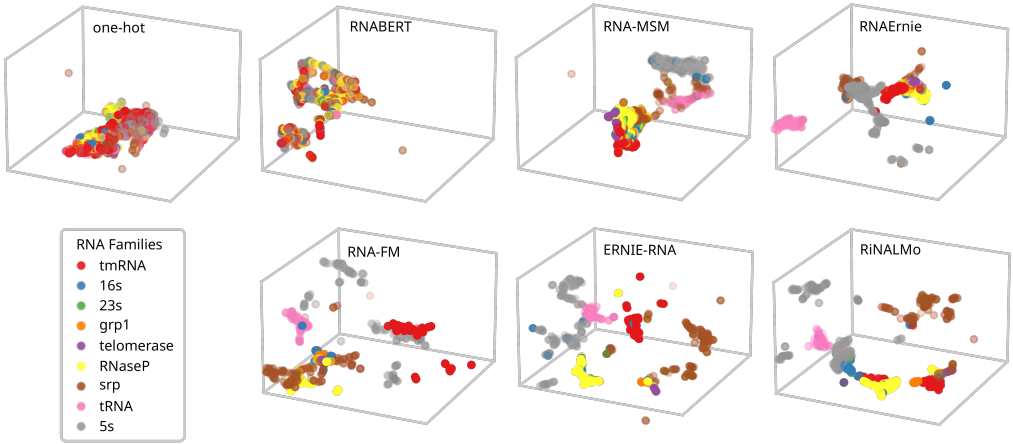
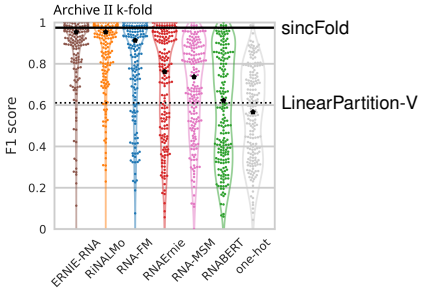


Results on datasets of increasing difficulty



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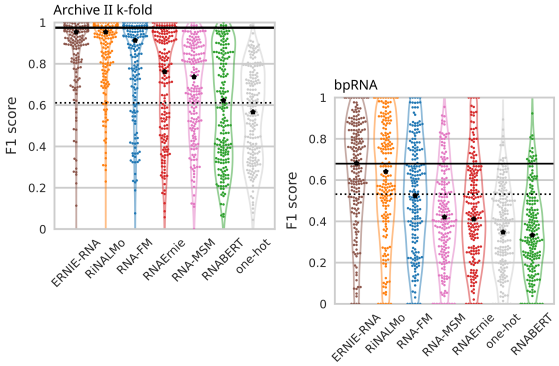

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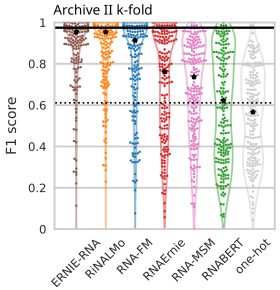

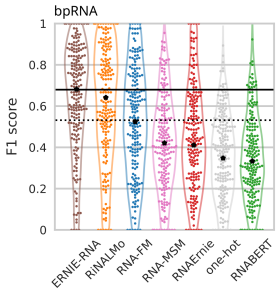

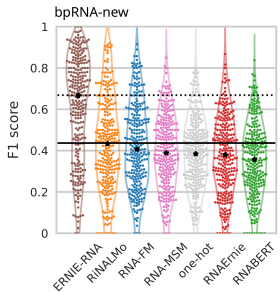

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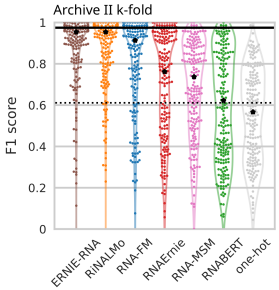
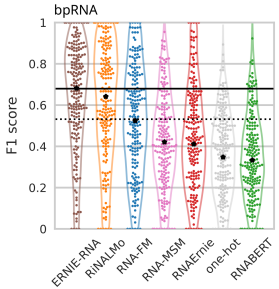
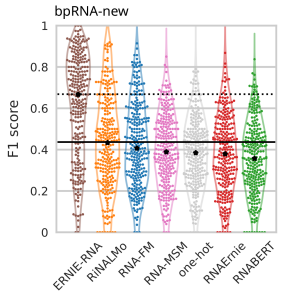
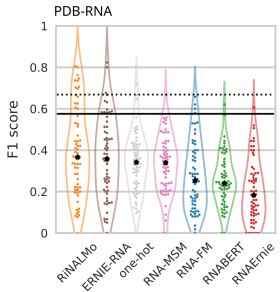
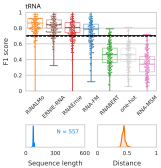


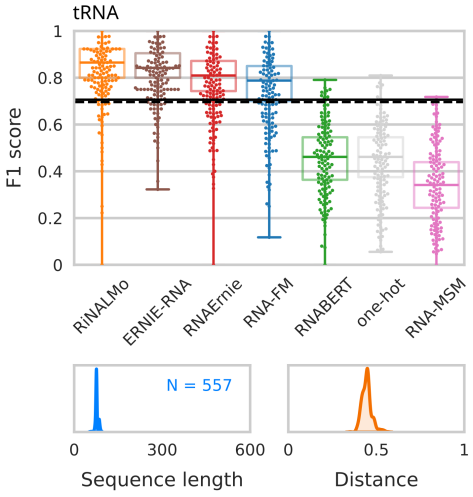






Across families: train on eight, test on the ninth





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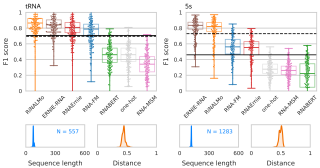

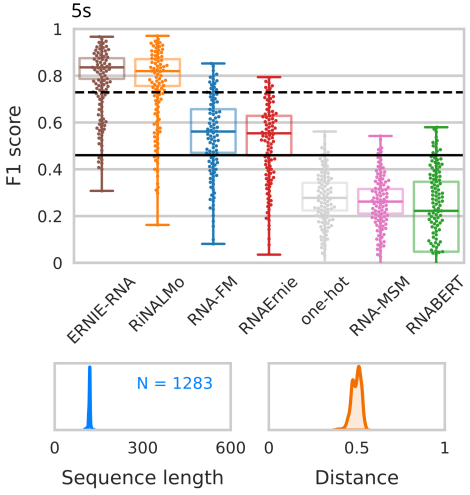

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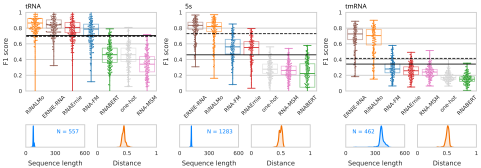

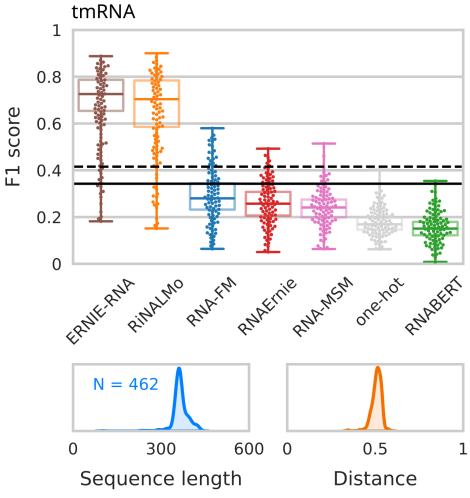

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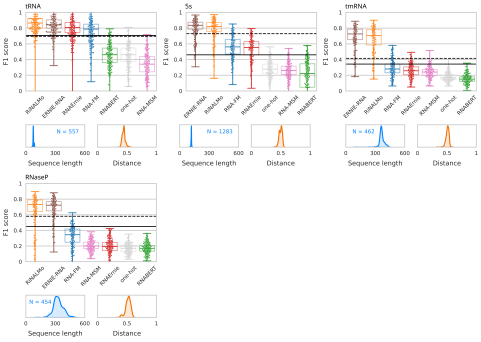

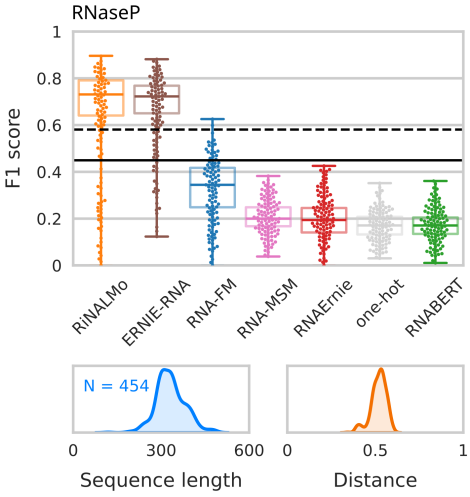

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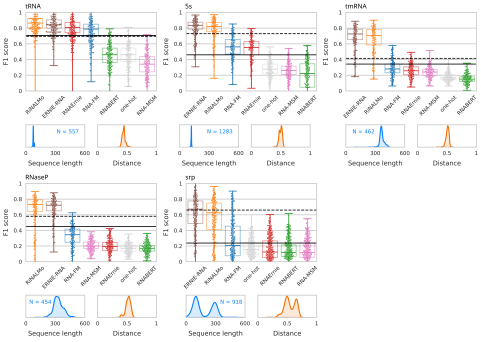

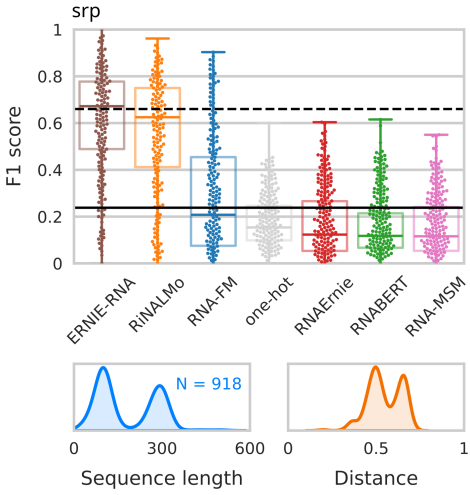

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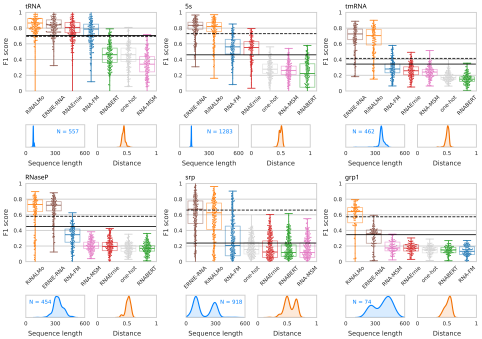

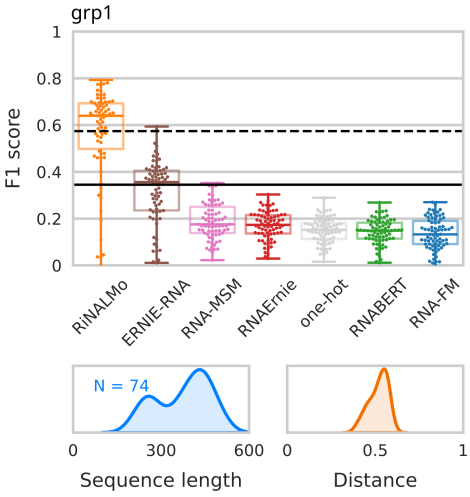

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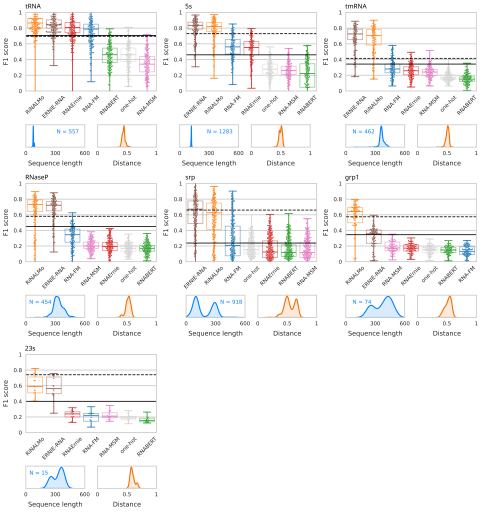

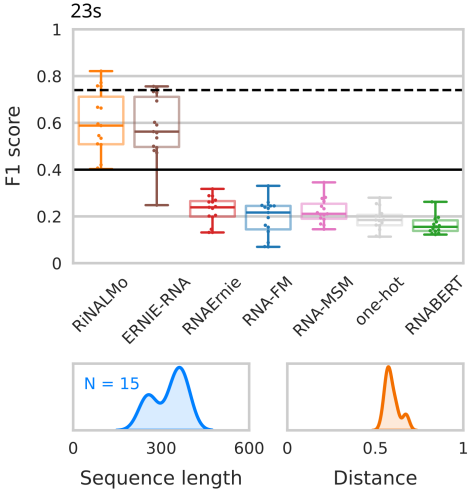

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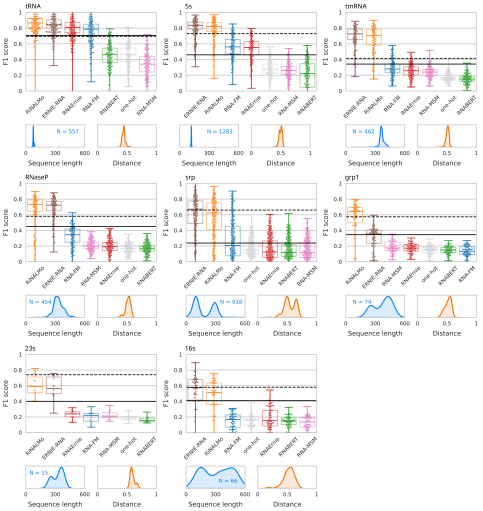

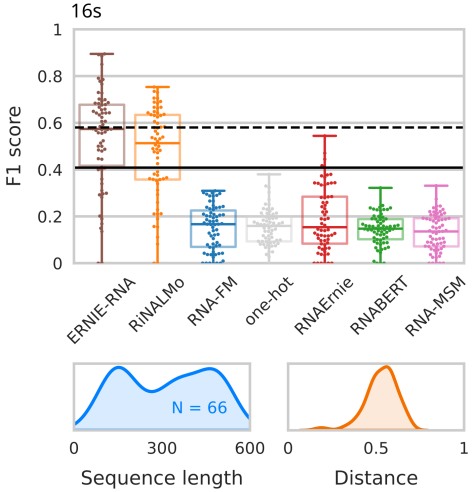

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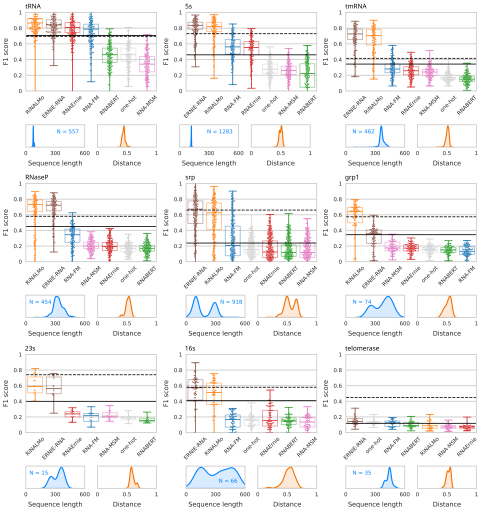

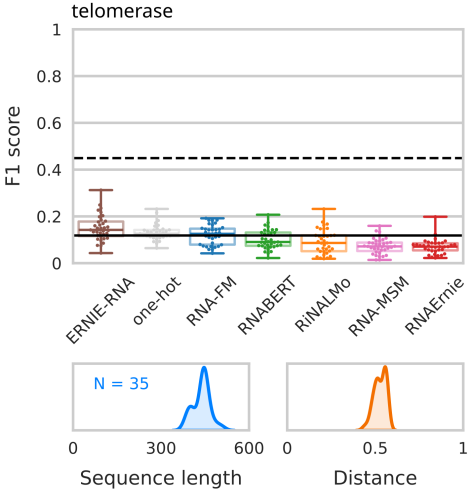

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LLMs clearly improve performance
. . . but which LLM is the best?
  - several protein LLMs, same classifier

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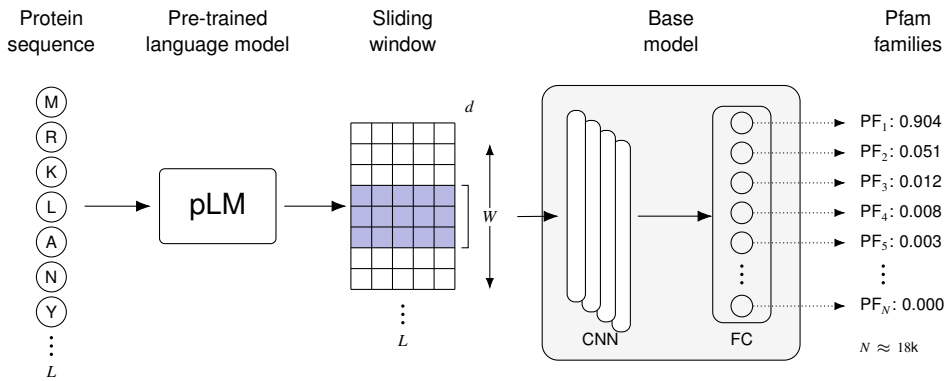
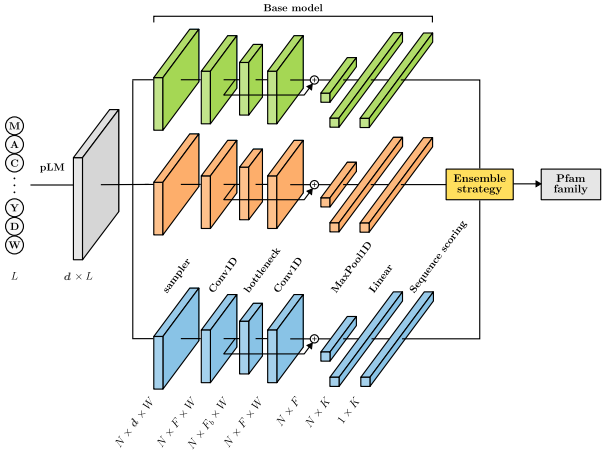
The base model



pLM: ESM-1b, ESM-1v, ESM2, ProtBert-BFD, ProtT5-XL-U50

ET-Pfam: an ensemble over embeddings



Embeddings:

- ESM-1b (2021)
  - —○ 1280-dim, ∼30M seqs, UniRef50
  - ↬ transformer (33), 650M params

—○ Vitale R, Bugnon LA, Fenoy E, Milone DH, Stegmayer G. Evaluating large language models for annotating proteins. Briefings in Bioinformatics 2024;25(3). doi:10.1093/bib/bbae177
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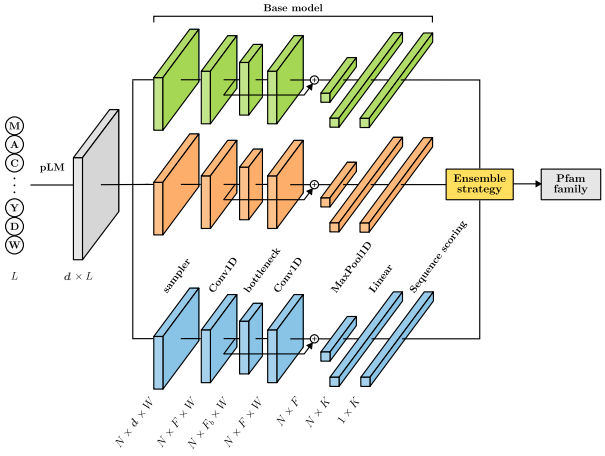

Embeddings:

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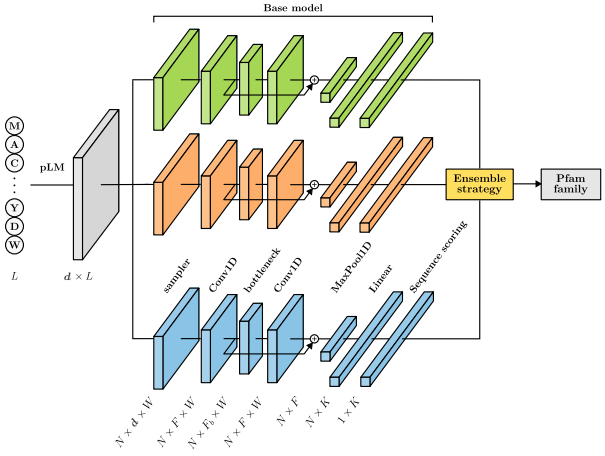

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- ESM-1v (2021)
  - —○ as ESM-1b, but UniRef90 and 5 models
- ESM2 (2023)
  - —○ 1280-dim, ∼65M seqs, UniRef50/90
  - ↬ transformer (33), 650M params

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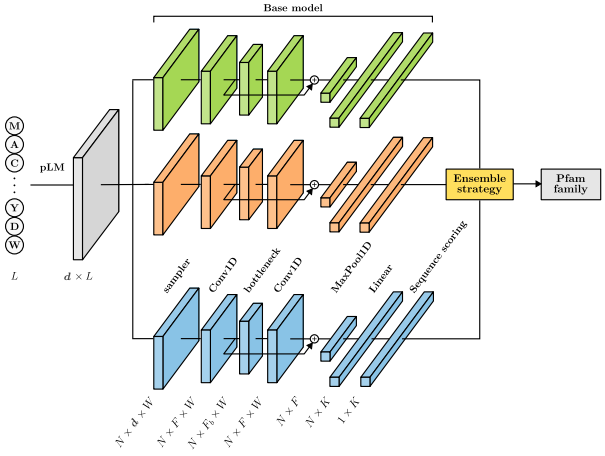

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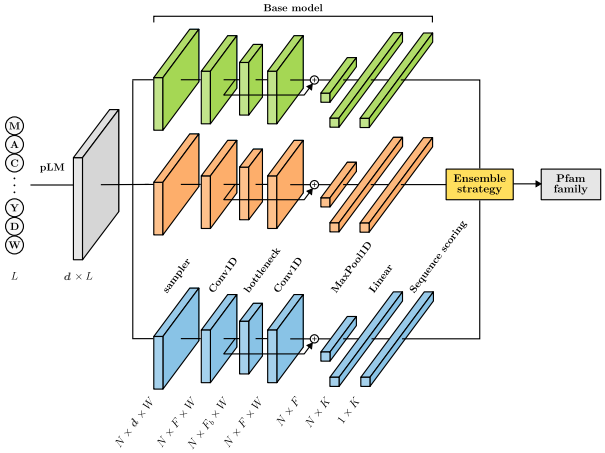
- ESM2 (2023)
  - —○ 1280-dim, ∼65M seqs, UniRef50/90
  - ↬ transformer (33), 650M params
- ProtBert-BFD (2022)
  - —○ 1024-dim, BFD
  - ↬ transformer (30), ∼420M params

—○ Vitale R, Bugnon LA, Fenoy E, Milone DH, Stegmayer G. Evaluating large language models for annotating proteins. Briefings in Bioinformatics 2024;25(3). doi:10.1093/bib/bbae177
—○ Duarte S, Vitale R, Escudero S, Fenoy E, Bugnon LA, Milone DH, Stegmayer G. ET-Pfam: ensemble transfer learning for protein family prediction. Bioinformatics 2026;42(4). doi:10.1093/bioinformatics/btag121

ET-Pfam: an ensemble over embeddings



Embeddings:

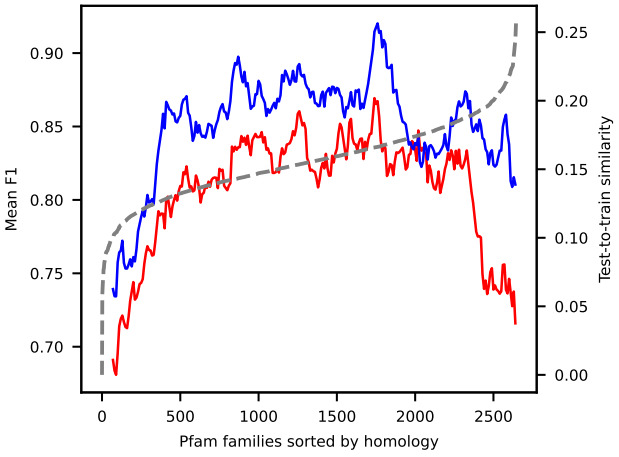
- ESM-1b (2021)
  - —○ 1280-dim, ∼30M seqs, UniRef50
  - ↬ transformer (33), 650M params
- ESM-1v (2021)
  - —○ as ESM-1b, but UniRef90 and 5 models
- ESM2 (2023)
  - —○ 1280-dim, ∼65M seqs, UniRef50/90
  - ↬ transformer (33), 650M params
- ProtBert-BFD (2022)
  - —○ 1024-dim, BFD
  - ↬ transformer (30), ∼420M params
- ProtT5-XL-U50 (2022)
  - —○ 1024-dim, 45M seqs, UniRef50
  - ↬ encoder (24), ∼1.2B of 3B params

—○ Vitale R, Bugnon LA, Fenoy E, Milone DH, Stegmayer G. Evaluating large language models for annotating proteins. Briefings in Bioinformatics 2024;25(3). doi:10.1093/bib/bbae177
—○ Duarte S, Vitale R, Escudero S, Fenoy E, Bugnon LA, Milone DH, Stegmayer G. ET-Pfam: ensemble transfer learning for protein family prediction. Bioinformatics 2026;42(4). doi:10.1093/bioinformatics/btag121

ET-Pfam: test errors on full Pfam

Input encoding	Models		Errors		
sequence	BLASTp			34.94%	
	pHMM			29.28%	
one-hot	ProtCNN			34.21%	
	ProtENN			18.05%	
ESM2	Single DL base model		13.94%	12.91%	12.98%
	ET-PFam ensemble	Simple voting	12.67%	11.77%	11.77%
		Score voting	12.34%	11.56%	11.45%
		LWM	12.28%	11.28%	11.41%
		LWF–perceptron	7.61%	7.32%	**7.00%**
		LWF–MLP	8.62%	9.76%	8.32%
ProtT5	Single DL base model		21.28%	15.92%	16.16%
ESM2-ProtT5	ET-PFam ensemble	Simple voting	13.44%	11.89%	11.89%
		Score voting	13.28%	12.16%	12.35%
		LWM	12.53%	11.69%	11.86%
		LWF–perceptron	17.30%	11.80%	9.59%
		LWF–MLP	47.28%	55.83%	50.30%
			CwS	SwA	SwC

Performance vs test-train similarity



What if the data is not a sequence?

–○ Fernandez Burda M, Bonazzola R, Valli AA, Castrillo G, Stegmayer G, Ferrante E, Milone DH. On the robustness of scRNA-seq foundation models for plant perturbation prediction. bioRxiv 2026. doi:10.64898/2026.08.21.746324 (under review)

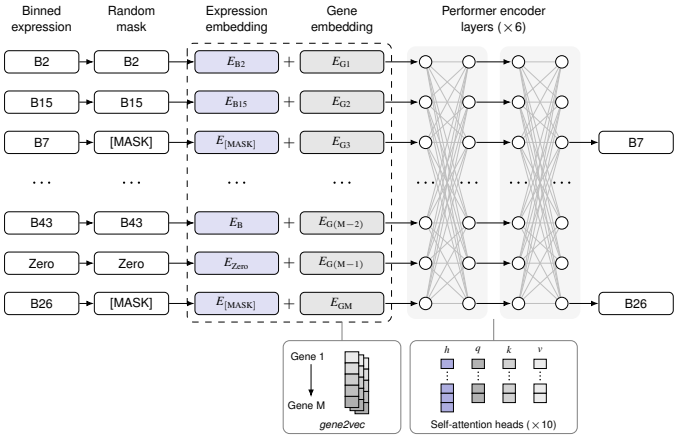
What if the data is not a sequence?

- Single-cell data is not a string of symbols
  - one expression value per gene

Fernandez Burda M, Bonazzola R, Valli AA, Castrillo G, Stegmayer G, Ferrante E, Milone DH. On the robustness of scRNA-seq foundation models for plant perturbation prediction. bioRxiv 2026. doi:10.64898/2026.08.21.746324 (under review)

What if the data is not a sequence?

- Single-cell data is not a string of symbols
  - ⊸ one expression value per gene
- Each gene is a token, each cell is a “sequence”
  - ⊸ as in scBERT: a 200-dim learned vector per gene
  - ↬ the normalized expression modulates that vector



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What if the data is not a sequence?

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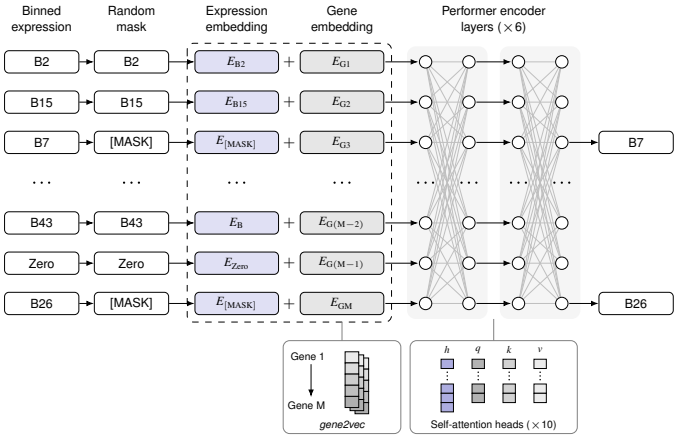
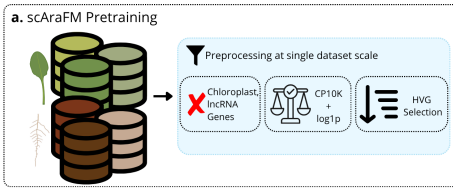
  - —∘ one expression value per gene
- Each gene is a token, each cell is a “sequence”
  - —∘ as in scBERT: a 200-dim learned vector per gene
  - ⇝ the normalized expression modulates that vector
- The vocabulary: highly variable genes
  - —∘ 4,397 in leaf, 4,054 in root



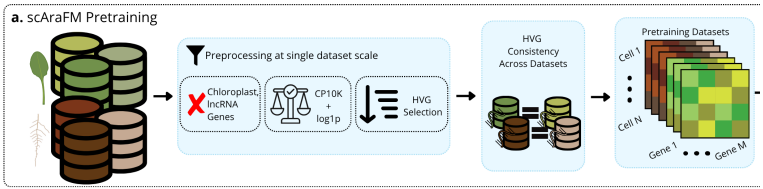
—∘ Fernandez Burda M, Bonazzola R, Valli AA, Castrillo G, Stegmayer G, Ferrante E, Milone DH. On the robustness of scRNA-seq foundation models for plant perturbation prediction. bioRxiv 2026. doi:10.64898/2026.08.21.746324 (under review)

Pretraining scAraFM



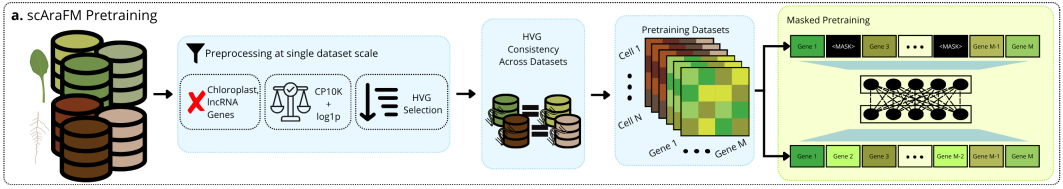
- Per study: harmonize gene IDs, normalize, log
  - organelle genes and lncRNA removed

Pretraining scAraFM



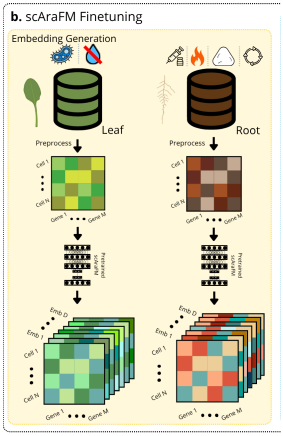
- Per study: harmonize gene IDs, normalize, log
  - organelle genes and lncRNA removed
- Shared highly variable genes across studies
  - 26 public *Arabidopsis thaliana* datasets
  - $\sim$17M cells in total

Pretraining scAraFM



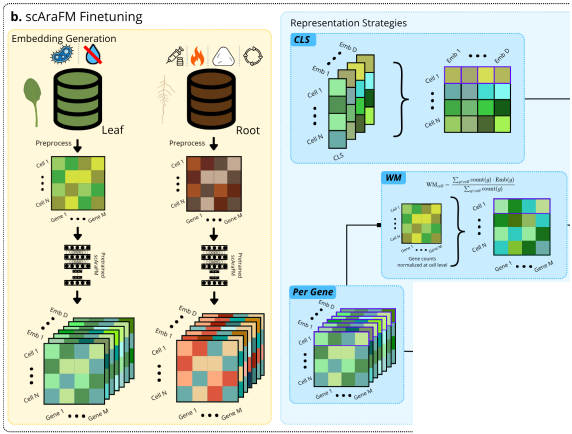
- Per study: harmonize gene IDs, normalize, log
  - organelle genes and lncRNA removed
- Shared highly variable genes across studies
  - 26 public *Arabidopsis thaliana* datasets
  - $\sim$17M cells in total
- Masked pre-training: hide genes, predict them

From embeddings to a prediction



Frozen model, embeddings for leaf and root

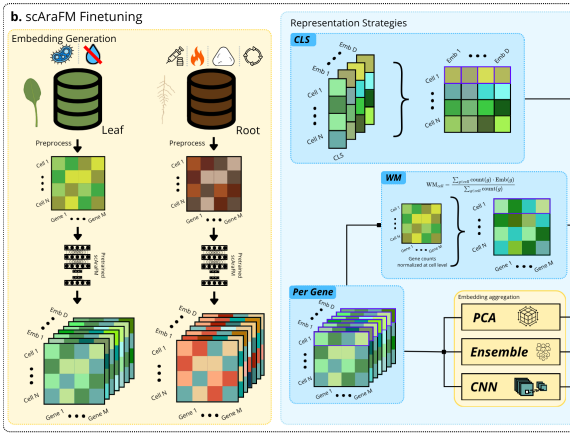
From embeddings to a prediction



Frozen model, embeddings for leaf and root

Compress the cell into one vector: CLS or weighted mean (WM)

From embeddings to a prediction

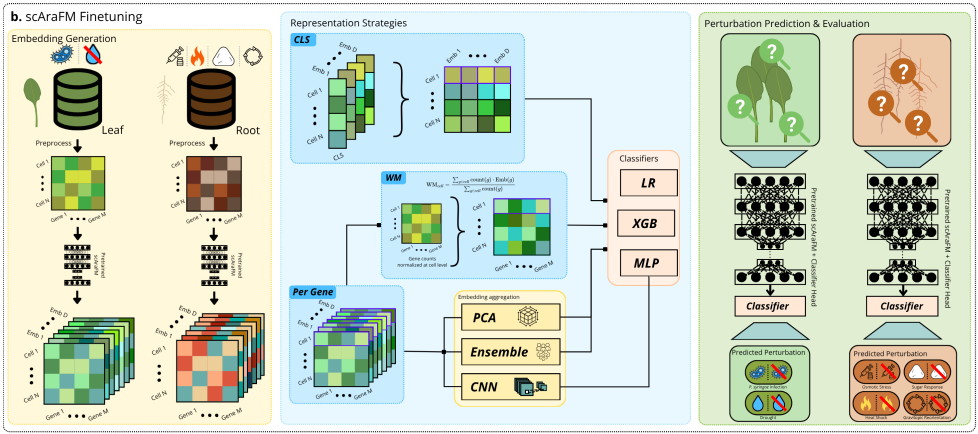


Frozen model, embeddings for leaf and root

Compress the cell into one vector: CLS or weighted mean (WM)

Keep each gene's identity: PCA, ensemble, or 1D CNN

From embeddings to a prediction



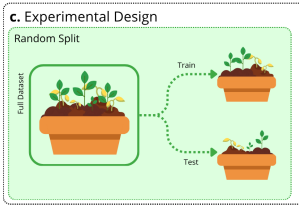
Frozen model, embeddings for leaf and root

Compress the cell into one vector: CLS or weighted mean (WM)

Keep each gene's identity: PCA, ensemble, or 1D CNN

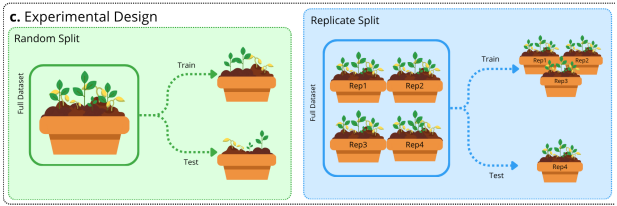
Three classifiers (LR, XGBoost, MLP) and a raw-count baseline

Three ways to split the data



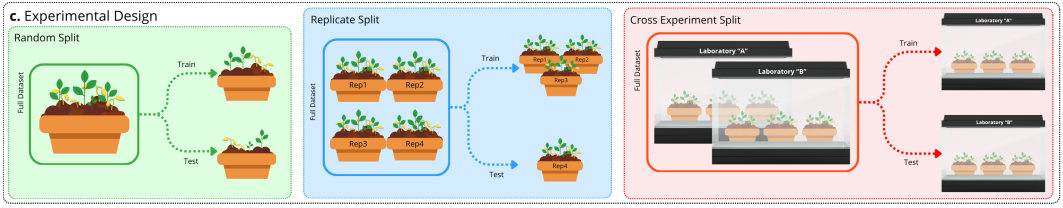
- Random split within one experiment

Three ways to split the data



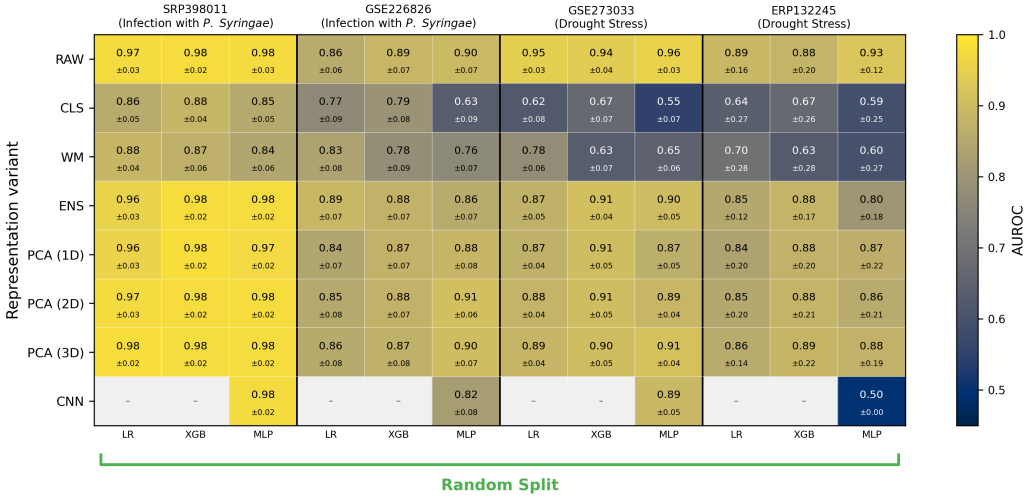
- Random split within one experiment
- Replicate split: one replicate held out

Three ways to split the data

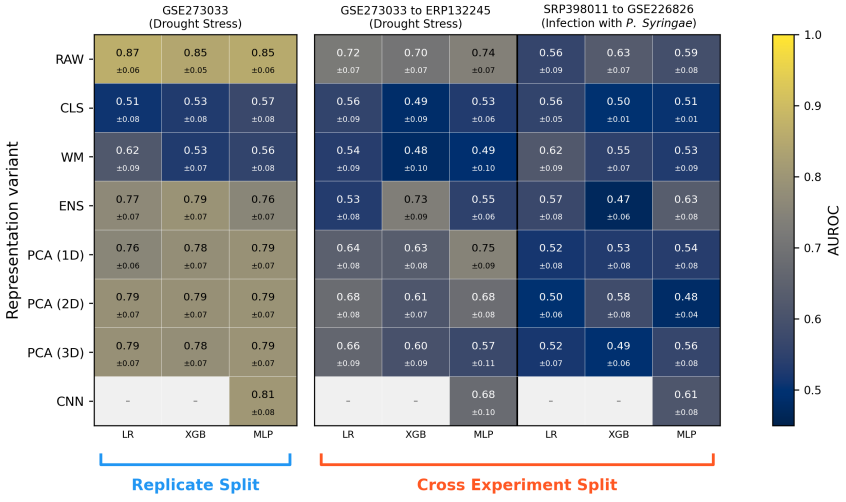


- Random split within one experiment
- Replicate split: one replicate held out
- Cross-experiment: train on one lab, test on another

Results


SRP398011
(Infection with P. Syringae)
GSE226826
(Infection with P. Syringae)
GSE273033
(Drought Stress)
ERP132245
(Drought Stress)
Representation variant
RAW
CLS
WM
ENS
PCA (1D)
PCA (2D)
PCA (3D)
CNN
LR
XGB
MLP
Random Split
AUROC
1.0
0.9
0.8
0.7
0.6
0.5

Results


GSE273033
(Drought Stress)
GSE273033 to ERP132245
(Drought Stress)
SRP398011 to GSE226826
(Infection with P. Syringae)
Representation variant
RAW
CLS
WM
ENS
PCA (1D)
PCA (2D)
PCA (3D)
CNN
0.87 ±0.06
0.85 ±0.05
0.85 ±0.06
0.72 ±0.07
0.70 ±0.07
0.74 ±0.07
0.56 ±0.09
0.63 ±0.07
0.59 ±0.08
0.51 ±0.08
0.53 ±0.08
0.57 ±0.08
0.56 ±0.09
0.49 ±0.09
0.53 ±0.06
0.56 ±0.05
0.50 ±0.01
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0.77 ±0.07
0.79 ±0.07
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0.47 ±0.06
0.63 ±0.08
0.76 ±0.06
0.78 ±0.07
0.79 ±0.07
0.64 ±0.08
0.63 ±0.08
0.75 ±0.09
0.52 ±0.08
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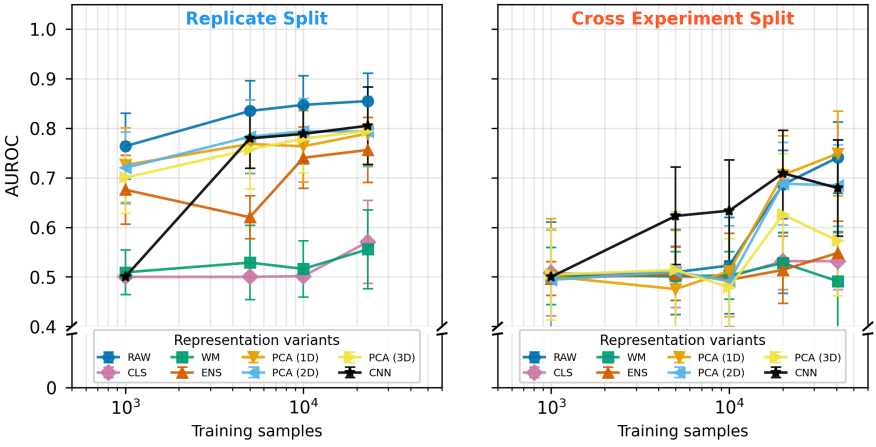
0.60 ±0.09
0.57 ±0.11
0.52 ±0.07
0.49 ±0.06
0.56 ±0.08
- - 0.81 ±0.08
- - 0.68 ±0.10
- - 0.61 ±0.08
LR XGB MLP LR XGB MLP LR XGB MLP
Replicate Split
Cross Experiment Split
AUROC
1.0
0.9
0.8
0.7
0.6
0.5

Results

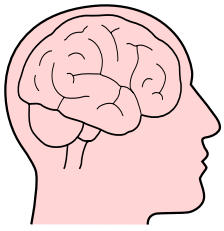

Replicate Split
Cross Experiment Split
AUROC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
10^3
10^4
Training samples
Representation variants
RAW
CLS
WM
ENS
PCA (1D)
PCA (2D)
PCA (3D)
CNN

Are we validating properly?

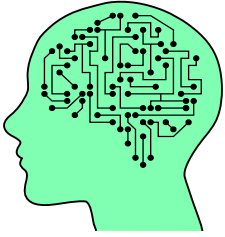
⊸ Bugnon LA, Kulemeyer G, Gerard M, Di Persia LE, Stegmayer G, Milone DH. Homology-aware cross-validation strategies for generalization assessment. bioRxiv 2026. doi:10.64898/2026.06.28.735057 (under review)

Human learning and machine learning

Theoretical models
(e.g. thermodynamics)



Data-driven models
(e.g. deep learning)



Human learning and machine learning

Theoretical models
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Data-driven models
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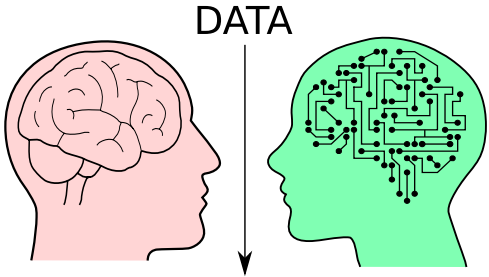
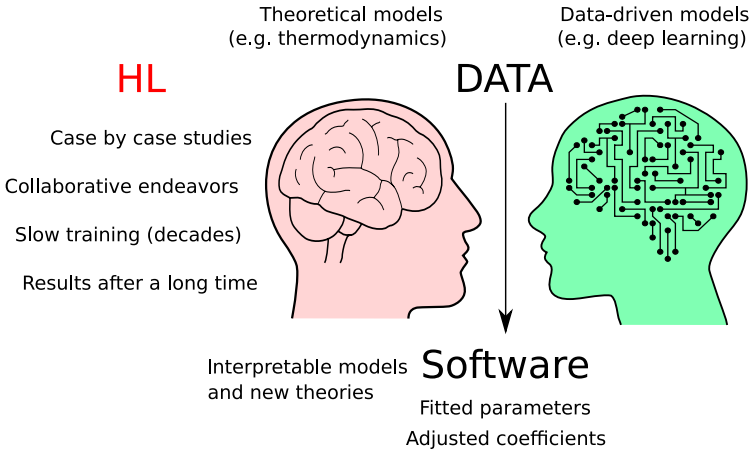


Software

Fitted parameters

Adjusted coefficients

Human learning and machine learning



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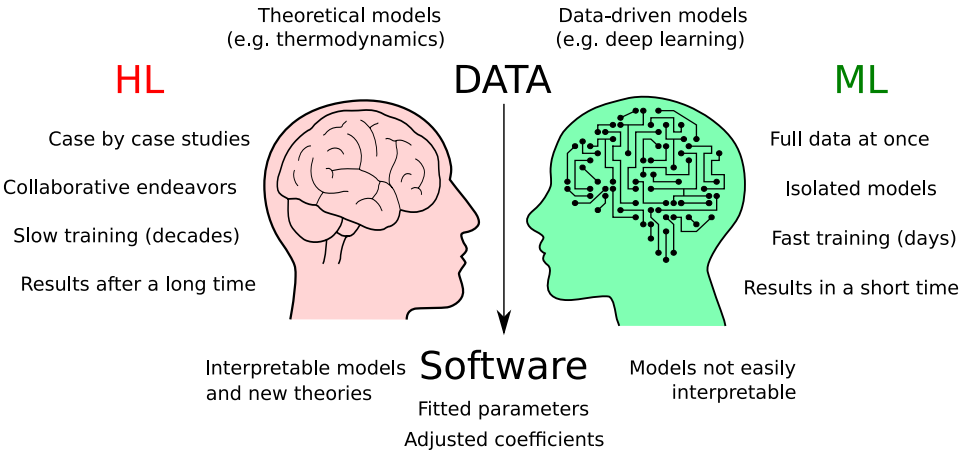
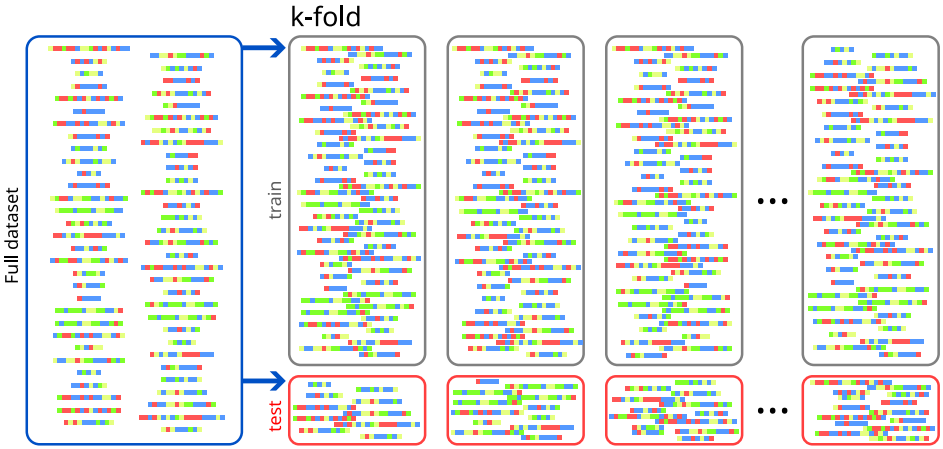
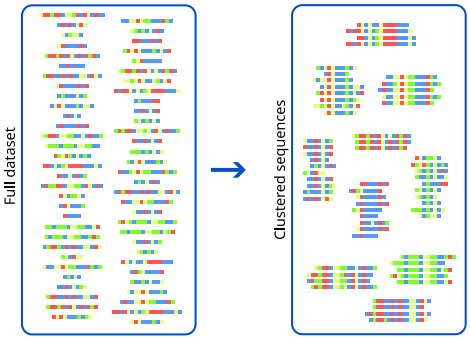


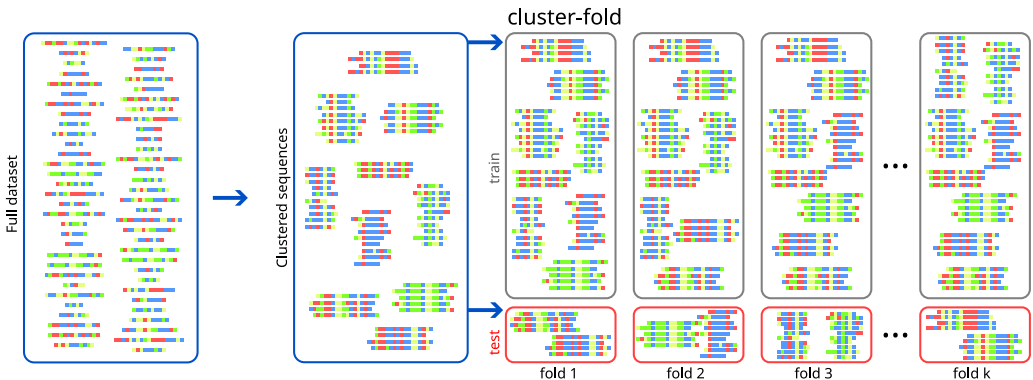
Random k-fold



Cluster-fold

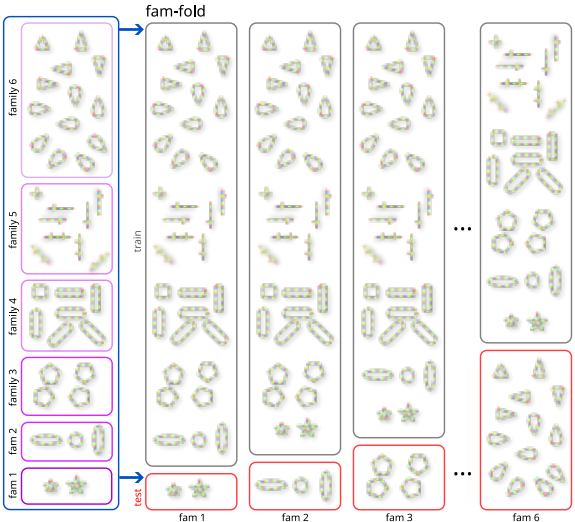


Cluster-fold

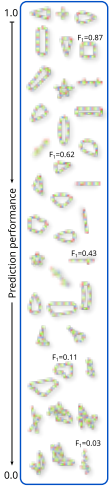


Family folds

Family folds



Human-learned folds



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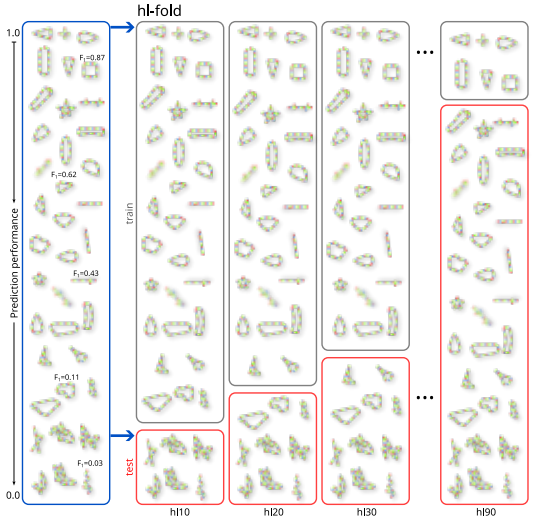
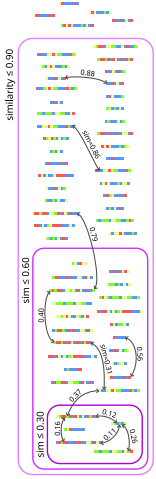


Similarity folds



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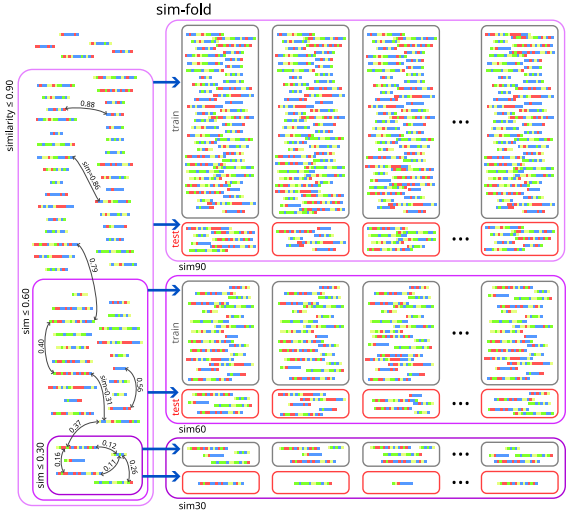
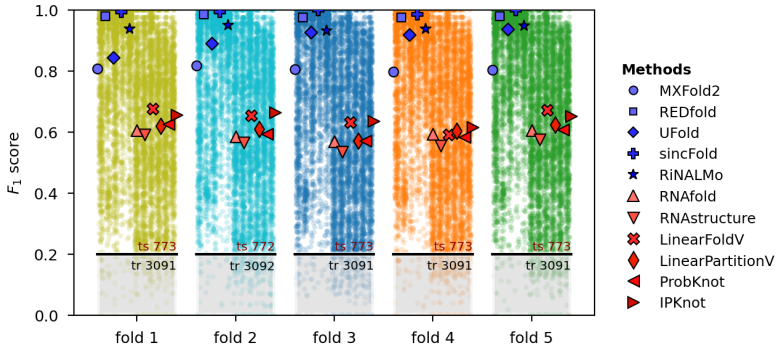
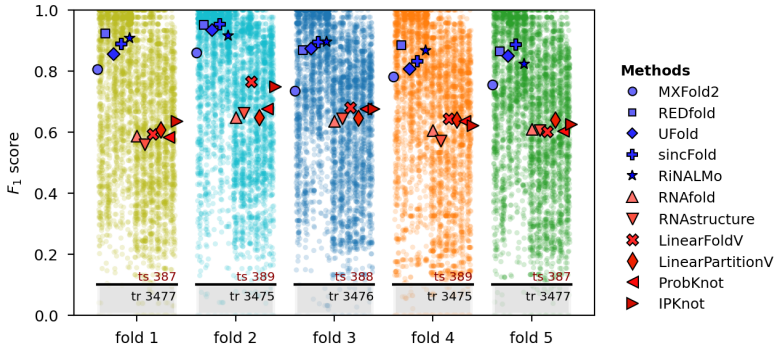


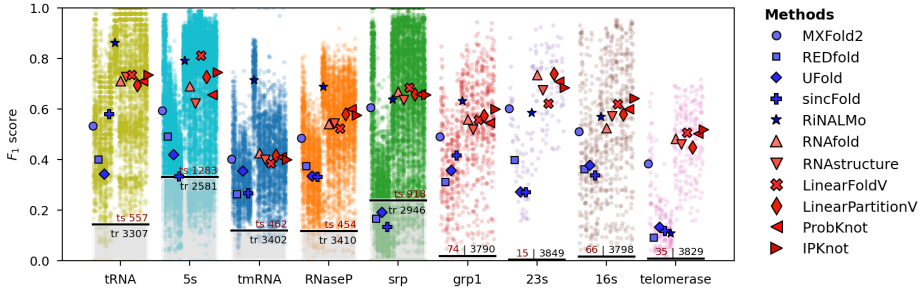
Results with random k-fold



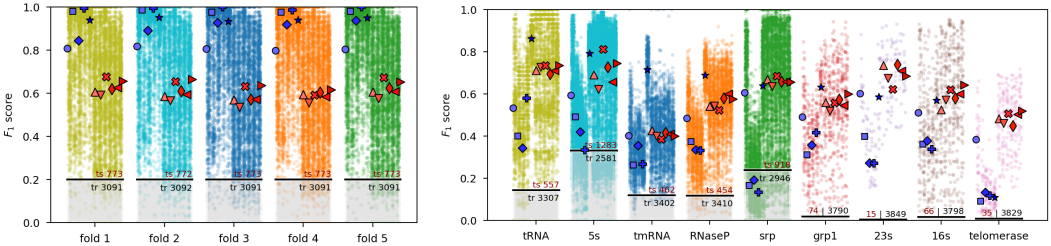
Results with cluster-fold



Results with family folds



Two winners, two leaks



Same methods, same data: only the split changed

The apparent winner is an artifact of the split

Data leakage vs **Knowledge** leakage

But... how hard is a family fold to learn?

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One counterexample is enough

- How far apart are the base pairs?
  - —○ in the reference structures, family by family

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  - for example, the tmRNA fold

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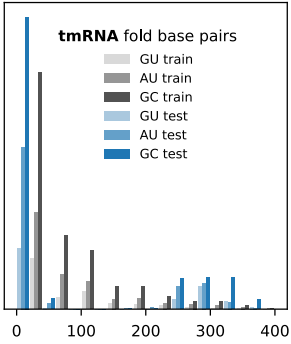
- One test family vs all the training families
  - for example, the tmRNA fold



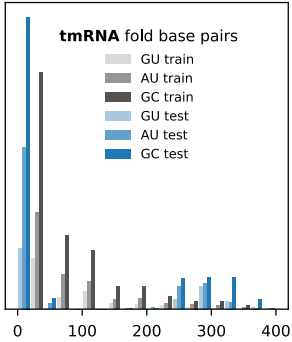
But... how hard is a family fold to learn?

One counterexample is enough

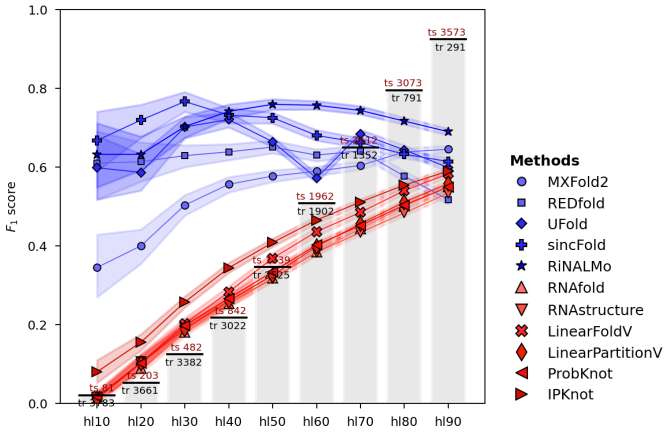
- How far apart are the base pairs?
  - in the reference structures, family by family
- A distribution of pair distances per family
  - estimated from the reference structures
- One test family vs all the training families
  - for example, the tmRNA fold

A task that cannot be learned

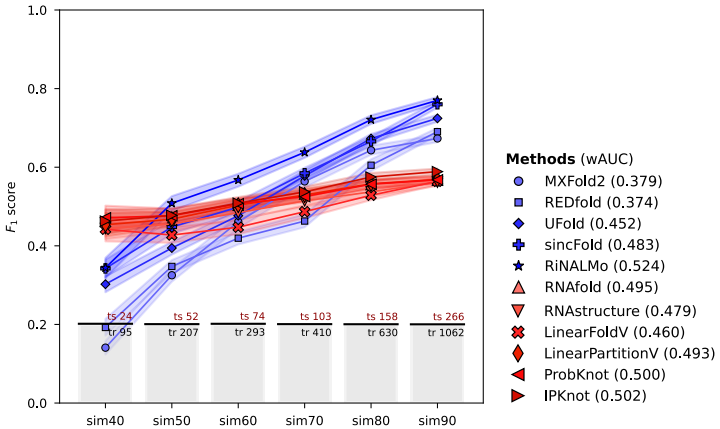
- tmRNA: base pairs at 250 to 400 nt
- Patterns that live only in the test family
- **Not a hard task: an impossible one!**



Results with human-learned folds



Results with similarity folds



Large Language Models for Biological Sequences

Do the models learn, or do they just memorize?

Gracias!

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03/11/26